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Bachelor's Programme "Applied Data Analysis"

UDC XXXXXXX

Research Project Report on the Topic:

Optimization of Pit Stop Strategies in Motorsport Using Telemetry and Simulation Data

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Annotation

This work addresses the problem of optimizing pit stop strategies in motorsports based on the analysis of telemetry and simulation data. Pit stop strategy is a key factor influencing race outcomes, as it determines tire usage patterns and the dynamics of positional interactions on the track.

The objective of the study is to develop a model for evaluating and comparing alternative pit stop strategies while accounting for tire degradation, race events, and changing race conditions. The research employs methods of time series analysis, optimization, and machine learning, as well as race simulation techniques.

The expected outcome of the work is a decision support algorithm that enables analysis of pit stop strategy effectiveness and investigation of the influence of various factors on strategic decisions. The results can be applied to race strategy analysis and further research in the field of intelligent motorsport strategy modeling.

Аннотация

Данная работа посвящена задаче оптимизации стратегий пит-стопов в автоспорте на основе анализа телеметрических и симуляционных данных. Стратегия пит-стопов является ключевым фактором, влияющим на результаты гонки, поскольку определяет характер использования шин и динамику позиционных взаимодействий на трассе.

Цель исследования заключается в разработке модели для оценки и сравнения альтернативных стратегий пит-стопов с учётом деградации шин, гоночных событий и изменяющихся условий гонки. В работе используются методы анализа временных рядов, оптимизации и машинного обучения, а также техники гоночного моделирования.

Ожидаемым результатом работы является алгоритм поддержки принятия решений, позволяющий анализировать эффективность стратегий пит-стопов и исследовать влияние различных факторов на стратегические решения. Полученные результаты могут быть применены для анализа гоночных стратегий и дальнейших исследований в области интеллектуального моделирования стратегий в автоспорте.

Keywords

- **Pit stop** - a scheduled stop during a race in which a car enters the pit lane to change tires and perform necessary technical servicing.
- **Pit stop strategy** - a set of decisions defining the number, timing, and characteristics of pit stops throughout a race in order to minimize total race time or maximize finishing position.
- **Tire degradation** - the gradual loss of tire performance caused by mechanical wear, thermal effects, and operating conditions, leading to reduced grip and slower lap times.
- **Race simulation** - a modeling approach that reproduces race dynamics, vehicle behavior, resource degradation, and the impact of strategic decisions within a virtual environment.
- **Machine learning** - a field of artificial intelligence focused on developing models that learn patterns from data and make predictions or decisions without explicit rule-based programming.
- **Time series analysis** - a collection of methods for analyzing temporally ordered data to identify trends, dependencies, and patterns, and to forecast future values.

Contents

1 Introduction

Modern motorsport represents a highly technological and dynamic environment in which race outcomes are determined not only by vehicle performance and driver skill, but also by the quality of strategic decision-making. One of the key components of race strategy is pit stop strategy, which defines the timing of tire changes, the selection of tire compounds, and the allocation of resources throughout the race distance. Errors in strategic choices can lead to significant losses in time and track position, whereas optimal decisions can provide a competitive advantage even when vehicles have comparable technical characteristics.

In recent years, advances in telemetry systems and data acquisition methods have resulted in the availability of large volumes of high-frequency data describing vehicle state, tire conditions, and race environment. At the same time, interest in applying data analysis, optimization, and machine learning techniques to support strategic decision-making in motorsport has increased. Nevertheless, the optimization of pit stop strategies remains a challenging problem due to the nonlinear nature of tire degradation, the influence of stochastic race events (such as safety car deployments and changing weather conditions), as well as the need to account for evolving track conditions and interactions with competitors.

The objective of this work is to develop a model for evaluating and comparing alternative pit stop strategies based on the analysis of telemetry and simulation data. Within the scope of the study, tire degradation, fuel mass, race pace dynamics, constructors, tire compounds, and the impact of external factors such as safety car periods, weather conditions, and traffic are examined. Based on these parameters, the applicability of optimization and machine learning methods for supporting strategic decision-making is investigated. The practical significance of the study lies in the potential application of the obtained results for analyzing the effectiveness of pit stop strategies and assessing the influence of various factors on overall race outcomes. The proposed approach can serve as a foundation for decision support systems in motorsport, as well as for further research in the field of intelligent race strategy modeling.

2 Literature Review

1) A comprehensive race simulation for AI strategy decision-making in motorsports. Max Boettinger, David Klotz, Mastering Nordschleife (2023)

Description. The authors present a high-fidelity race simulation platform designed to support strategic decision-making in motorsports using artificial intelligence methods. A comprehensive simulator of the Nürburgring Nordschleife circuit is developed, modeling vehicle dynamics, tire behavior, fuel consumption, and race events under long-duration racing conditions.

Connection to the topic. Simulation-based modeling plays a key role in research on pit stop strategy optimization, as real-world experimentation with alternative strategies is constrained by high costs and practical limitations.

Methods and models.

- Physics-based race simulation incorporating detailed modeling of vehicle dynamics and track-specific characteristics.
- Tire and fuel models describing performance degradation and resource consumption over time.
- Discrete-event modeling of race processes, including pit stops, lap progression, and strategic decision-making.
- Integration of artificial intelligence and reinforcement learning methods for evaluating and optimizing strategic policies.

Results.

- The developed simulator reproduces realistic lap times, degradation trajectories, and race dynamics.
- Strategies based on AI methods demonstrate higher stability and effectiveness compared to baseline or heuristic strategies.

2) Explainable Time Series Prediction of Tyre Energy in Formula One Race Strategy. Jamie Todd, Junqi Jiang, Aaron Russo, Steffen Winkler, Stuart Sale, Joseph McMillan, Antonio Rago. (2025)

Description. The article addresses the problem of predicting tire energy in Formula 1 using time series analysis and explainable machine learning methods. The authors focus on modeling the dynamics of tire energy as a latent indicator reflecting grip conditions and degradation, based on telemetry and race data.

Connection to the topic. Accurate and interpretable prediction of tire state over time enables the assessment of expected performance degradation within a stint and supports the selection of an optimal timing for tire changes. The use of explainable time series models is particularly important in the context of strategic modeling, where decisions must be not only optimal but also justified in terms of physical and race-related factors. Such an approach can be integrated into simulation and optimization frameworks for pit stop strategies to increase trust in predictions and improve the quality of strategic decision-making.

Methods and models.

- Time series forecasting models for estimating tire energy dynamics over a race stint.
- Application of explainable machine learning methods to interpret the influence of input features on predictions.
- Analysis of Formula 1 telemetry and race data, including tire parameters, pace, and session conditions.
- Use of feature attribution mechanisms to identify key factors driving changes in tire energy.

Results.

- The proposed models demonstrate high accuracy in predicting tire energy across multiple forecasting horizons.
- Explainability methods enable the identification of the main factors influencing tire energy degradation and recovery during a race.
- Interpretable time series models can effectively support strategic decision-making in Formula 1 by balancing predictive accuracy and model transparency.

3) Optimizing pit stop strategies in Formula 1 with dynamic programming and game theory. Felipe Aguad, Charles Thraves. (2024)

Description. The article addresses the optimization of pit stop strategies in the context of direct competition between two drivers. The strategic problem is formulated as a zero-sum feedback Stackelberg game, in which drivers decide at each lap whether to stay on track or perform a pit stop with a specific tire compound choice. The model accounts for the impact of uncertain events, such as yellow flags, as well as stochastic lap times, and provides an algorithm for computing the strategic equilibrium.

Connection to the topic. Optimizing pit stop strategies requires consideration not only of internal factors (tire degradation, race pace, and pit stop duration) but also of interactions with competitors, which can significantly alter the value of different strategic choices. Formulating the problem as a dynamic game enables formal analysis and computation of optimal decisions under competitive conditions. This approach extends traditional optimization models by incorporating strategic interaction and uncertainty, thereby better reflecting real-world motorsport environments.

Methods and models.

- Dynamic Programming is used to sequentially determine pit stop decisions at each lap.
- A zero-sum feedback Stackelberg game is formulated between two drivers, where the leader and the follower make decisions at each time step.
- Initial tire compound choices, lap-by-lap strategies, and uncertain events such as yellow flags and lap time variability are explicitly considered.
- An algorithm is provided to compute the Stackelberg equilibrium using backward induction and dynamic programming.

Results.

- The existence of a feedback Stackelberg equilibrium for pit stop strategies in a two-driver setting is established.
- Strategies that account for competitive interactions differ substantially from those that ignore competitors; a driver adopting a strategically optimal policy can increase winning probability by more than 15% compared to a strategy that does not consider the opponent.
- The model captures the impact of uncertain events, such as Safety Car or Virtual Safety Car periods, on optimal pit stop decisions.

4) Model Predictive Control Strategies for Electric Endurance Race Cars Accounting for Competitors' Interactions. Jorn van Kampen, Mauro Moriggi, Francesco Braghin, Mauro Salazar. (2024)

Description. The article examines the application of Model Predictive Control (MPC) methods for strategic planning in electric endurance racing. The authors develop a control model that accounts not only for the dynamics of the ego vehicle but also for interactions with competitors on the track. Particular attention is given to the joint management of energy consumption, race pace, and strategic decisions over long planning horizons.

Connection to the topic. Accounting for interactions with competitors is a critical aspect of real-world race strategies, as decisions regarding pace and pit stops directly depend on the relative positions of vehicles and the anticipated behavior of rivals. The use of predictive control allows strategic decisions to be formulated as a constrained optimization problem, where the strategy is updated as new information about the race state becomes available. This approach is applicable to simulation-based pit stop strategy models, particularly in scenarios that require consideration of uncertainty, resource dynamics, and the influence of external agents.

Methods and models.

- Model Predictive Control (MPC) for optimizing race strategy over a finite planning horizon.
- A dynamic model of an electric race car, including energy consumption, battery constraints, and vehicle dynamics.
- Modeling of interactions with competitors through behavior prediction and incorporation of relative states into the optimization problem.
- Formulation of a constrained optimal control problem reflecting race regulations, physical limits, and strategic objectives.

Results.

- The proposed MPC-based strategies demonstrate improved race performance compared to baseline strategies that do not account for interactions with competitors.
- Considering competitor behavior enables more effective management of race pace and energy consumption over long distances.
- The results indicate that predictive control is a promising tool for strategic planning in endurance racing, particularly in multi-agent and dynamically evolving environments.

5) A State-Space Approach to Modeling Tire Degradation in Formula 1 Racing. Cole Cappello, Andrew Hoegh. (2025)

Description. The article focuses on the development of a Bayesian stochastic model for tire degradation in Formula 1. The authors propose a state-space approach in which the latent state the tire degradation rate is inferred from lap time data and fuel mass. The model also accounts for pit stops as state resets, while lap time is modeled as a function of the current tire wear state and fuel load.

Connection to the topic. Tire degradation is one of the key factors determining race pace and the timing of strategic decisions. Accurate modeling of tire wear enables more reliable prediction of lap time evolution over a stint and facilitates the evaluation of the trade-off between continuing on the current tire set and executing a pit stop. The use of a probabilistic state-space model provides a dynamic representation of the degradation process and allows uncertainty arising from fuel effects, driver behavior, and stochastic race events to be explicitly accounted for. Such an approach can be integrated into simulation and optimization frameworks for race strategy, where sequential state updates and forecasting the consequences of alternative pit stop strategies are required.

Methods and models.

- State-space model: the system defines a latent state representing the tire degradation rate and an observed variable (lap time), linked through probabilistic equations.
- Bayesian approach: model parameters are estimated via posterior distributions, allowing uncertainty to be explicitly incorporated.
- Fuel mass as a regressor affecting lap time; pit stops as state resets reflecting tire changes.
- Model extensions: compound-specific degradation to capture differences between tire compounds, and a skewed-t observation model to approximate asymmetric driver-related errors.

Results.

- The state-space model outperforms a traditional ARIMA approach in predicting lap times while accounting for tire degradation.
- The variant with a skewed-t observation model is the most stable and accurate, highlighting the importance of modeling asymmetric error distributions.

- Differences between tire compounds were not statistically significant in the presented case study (Austrian Grand Prix 2025); however, the model structure allows for extension to multiple races and drivers.

- The proposed approach is interpretable, computationally efficient, and easily scalable, making it potentially suitable for real-time strategic modeling.

6) On the optimization of pit stop strategies via dynamic programming. Oscar F. Carrasco Heine, Charles Thraves. (2022)

Description. The article proposes an algorithm for selecting optimal pit stop laps based on dynamic programming. The main contribution of the work is that the race strategy is computed directly from a model of tire degradation and pit stop time, without relying on simulations or heuristic rules.

Connection to the topic. The proposed approach provides a mathematical foundation for algorithmic optimization of pit stop strategies and can be used as a basic component of more advanced models, including those that account for stochastic events, interactions with competitors, or race simulation frameworks.

Methods and models.

- The pit stop strategy problem is formulated as a dynamic programming problem with discrete time, where each time step corresponds to a race lap.

- The system state includes information about the current lap, tire condition, and previous pit stop decisions.

- The cost function represents total race time and accounts for: tire degradation and its effect on lap time; a fixed time loss associated with performing a pit stop; constraints on the allowable number of pit stops.

- The optimal strategy is computed using backward induction, which determines the optimal decision (pit stop or stay on track) at each stage of the race.

Results.

- Dynamic programming can effectively identify optimal pit stop strategies across different race scenarios.

- The resulting strategies differ substantially from heuristic or predefined approaches, particularly in cases with pronounced tire degradation.
- Accurate modeling of tire degradation and pit stop time losses is critical for minimizing total race time.
- The proposed model can be easily extended to incorporate stochastic factors and competitive interactions, enabling further research.

7) Explainable Reinforcement Learning for Formula One Race Strategy. Devin Thomas, Junqi Jiang, Avinash Kori, Aaron Russo, Steffen Winkler, Stuart Sale, Joseph McMillan, Francesco Belardinelli, Antonio Rago. (2025)

Description. The article proposes an approach to Formula 1 race strategy selection using reinforcement learning trained in a race simulation environment. The authors introduce the RSRL (Race Strategy Reinforcement Learning) model as an alternative to traditional rule-based strategies and Monte Carlo scenario enumeration. A distinguishing feature of the work is the combination of reinforcement learning with a set of explainability methods.

Connection to the topic. The goal of the study is to address race strategy as a sequential decision-making problem solved by an RL agent that can adapt to changing race conditions, such as the deployment of a Safety Car. An important contribution is the focus on real-time applicability and model trust, as the authors demonstrate how a race strategist can interpret the agent's decisions.

Methods and models.

- The race strategy selection problem is formulated as a reinforcement learning task with discrete decisions made at each lap of the race. The action space includes continuing the stint or performing a pit stop with a specific tire compound choice.
- A deep neural network with a recurrent architecture is used to approximate the policy, allowing the model to capture temporal dynamics and partial observability of the race process.
- The environment state includes aggregated telemetry and strategic features, such as the current tire compound, tire degradation level, race progress, time gaps between cars, and Safety Car status.

- Training is performed in a simulation environment representing race scenarios, while decision interpretability is achieved through explainable machine learning methods applied to the learned policy.

Results.

- Strategies obtained via reinforcement learning outperform baseline heuristic approaches in terms of overall race outcome in the simulation environment.
- The model demonstrates the ability to adapt its strategy to changing race conditions, including the presence of Safety Car periods.
- The use of explainability methods enables interpretation of the agent's decisions and identification of key factors influencing pit stop timing and tire compound selection.

8) Towards Learning-Based Formula 1 Race Strategies. Giona Fieni, Joschua Wüthrich, Marc-Philippe Neumann, Mohammad H. Moradi, Christopher H. Onder. (2025)

Description. The article proposes a systematic approach to the development of Formula 1 race strategies based on machine learning and optimization methods. The authors view race strategy as the result of jointly solving pace selection, resource management, and pit stop decision-making problems, formulating it as a multi-level optimization problem. The core idea of the work is a shift from rigid, rule-based strategies toward learnable models capable of adapting to different race conditions.

Connection to the topic. In contrast to approaches that focus exclusively on local pit stop timing decisions, this study investigates race strategy in a broader context, where pit stop decisions are coordinated with pace management and resource allocation. The proposed learning-based approach is relevant for simulation-based strategy models, as it enables analysis of how optimal pit stop decisions vary with race state and long-term objectives, and it serves as a bridge between classical optimization methods and data-driven strategies.

Methods and models.

- Race strategy is formulated as a multi-level optimization problem that integrates pit stop selection, pace control, and resource allocation over the entire race.
- Learning-based methods are combined with optimization models, allowing strategies to be trained on simulation data and applied to new race scenarios.

- The model relies on physically interpretable representations of race state, including tire characteristics, race progress, and regulatory constraints.
- A modular architecture is emphasized, enabling integration of the strategic layer (pit stops and pace) with lower-level vehicle and track dynamics models.

Results.

- Learning-based strategies can reproduce and, in some cases, outperform traditional rule-based or manually designed strategies in a simulation environment.
- Combining learning and optimization leads to more coherent strategies, where pit stop decisions are logically aligned with the overall race plan.
- The findings highlight the potential of learnable models as a foundation for further development of adaptive and multi-agent race strategies in motorsports.

9) Game Theory in Formula 1: Multi-agent Physical and Strategical Interactions. Giona Fieni, Marc-Philippe Neumann, Francesca Furia, Alessandro Caucino, Alberto Cerofolini, Vittorio Ravaglioli, Christopher H. Onder. (2025)

Description. The article investigates race strategy planning in Formula 1 under multi-agent interactions, where the decisions of one driver affect the dynamics and outcomes of others. The authors model a race as a multi-agent system in which physical interactions (pace, overtaking, time gaps) and strategic decisions (pit stops, tire selection, race pace) are formulated within a game-theoretic framework. The key contribution of the work lies in integrating strategic decision-making with a physics-based race model within a unified game-theoretic formulation.

Connection to the topic. Optimizing pit stop strategies in real races requires accounting for competitor behavior, as relative track position, traffic, and anticipated opponent decisions significantly influence strategy effectiveness. In this work, pit stops are treated not as isolated decisions but as components of strategic interactions among multiple agents, making the approach highly relevant to competitive strategy optimization. By applying game theory, the authors explicitly model how the strategic choices of one participant affect the optimal responses of others, extending classical single-agent pit stop optimization models and bringing the formulation closer to real-world racing conditions.

Methods and models.

- The race is formulated as a multi-agent dynamic game in which each agent makes decisions regarding race pace and strategic actions, including pit stops.
- Physics-based models of vehicle dynamics and tire degradation are used to describe the evolution of race states.
- Strategic decisions are modeled within a game-theoretic framework that captures interdependencies between agents and their impact on future system states.
- Equilibrium concepts are employed to analyze stable strategies in a competitive multi-agent environment.

Results.

- Optimal pit stop and pace strategies change significantly when interactions between drivers are taken into account, compared to single-agent models.
- Game-theoretic models can identify scenarios in which aggressive or conservative strategies become optimal depending on competitor behavior.
- Multi-agent game-theoretic approaches are a promising direction for modeling race strategies in Formula 1, particularly for analyzing competitive effects and tactical decision-making.

10) Data-driven pit stop decision support for Formula 1 using deep learning models. Abhijai Sasikumar, A. Anny Leema, P. Balakrishan. (2025)

Description. The article proposes a data-driven approach to pit stop decision support in Formula 1 based on deep learning methods. The authors formulate pit stop timing as a prediction and classification problem using historical race and telemetry data. Unlike optimization- or simulation-based approaches, the study focuses on extracting patterns directly from data to predict strategic decisions.

Connection to the topic. The study proposes a data-driven approach as an alternative to traditional model-based methods. Instead of explicitly defining tire degradation models or race dynamics, the authors employ deep neural networks to learn the relationships between race state variables and optimal strategic decisions. As a result, the work complements optimization- and reinforcement learning-based approaches by demonstrating how pit stop strategies can be derived directly from data without relying on explicit physical models.

Methods and models.

- The pit stop decision support problem is formulated as a supervised machine learning task using historical Formula 1 race data.
- Deep learning models are employed to predict optimal pit stop timing based on input features describing the race state.
- Input data include race pace indicators, tire-related variables, track position, and contextual race information.
- The models are trained in a supervised setting, learning to reproduce strategic decisions observed in historical race data.

Results.

- The proposed deep learning models achieve high accuracy in predicting pit stop decisions based on race state information.
- A data-driven approach can effectively serve as a decision support system for race strategists.
- Prediction performance depends on the volume and representativeness of the training data, highlighting the importance of access to detailed telemetry.

11) Virtual Strategy Engineer: Using Artificial Neural Networks for Making Race Strategy Decisions in Circuit Motorsport. Alexander Heilmeier, Andre Thomaser, Michael Graf, Johannes Betz. (2020)

Description. The article introduces the concept of a Virtual Strategy Engineer - a decision support system for circuit motorsport strategy based on artificial neural networks. Race strategy, including pit stop timing and tire selection, is treated as an approximation of optimal strategic behavior learned from historical data and race simulations. The main contribution of the work lies in using neural networks to emulate professional race strategist decisions in near real-time.

Connection to the topic. Pit stop strategies are modeled as a direct mapping from race state to strategic decisions without an explicit optimization procedure. This approach complements model-based and reinforcement learning methods by enabling strategies to be derived directly from data and applied under strict computational time constraints.

Methods and models.

- The strategic problem is formulated as a mapping from race state to strategic decisions, approximated by an artificial neural network.

- Model training is performed in a supervised setting using data generated by a race simulator, where target strategies are obtained from conventional simulation-based and analytical methods.

- The neural network acts as a functional approximator of race strategy, replacing computationally expensive scenario enumeration and repeated simulations during the race.

- The model architecture is designed for real-time application, producing strategy recommendations through a single forward pass without iterative optimization.

- The approach allows scaling to different tracks and race scenarios through retraining on extended sets of simulated data.

Results.

- The neural network reproduces strategic decisions comparable to those obtained with traditional simulation-based approaches.

- The model demonstrates suitability for decision support with low inference latency, which is critical for real-time race applications.

- Good scalability across different circuits and race scenarios is reported given sufficient training data.

12) Planning Formula One race strategies using discrete-event simulation. J. Bekker, W. Lotz. (2009)

Description. The article proposes an approach to Formula 1 race strategy planning based on discrete-event simulation. The authors model a race as a sequence of events - such as lap completions, pit stops, overtakes, and incidents—allowing strategic dynamics to be analyzed at the race level. The work represents one of the early systematic attempts to formalize Formula 1 race strategy within an operations research framework.

Connection to the topic. Pit stop strategies are treated as key events within the discrete structure of a race, directly influencing the final outcome. This representation enables analysis of the consequences of different strategic decisions within an integrated race model, where pit stops interact with pace, traffic, and stochastic events. The discrete-event formulation provides a foundation for simulation-based strategy analysis and serves as a precursor to more recent approaches that employ optimization, reinforcement learning, and data-driven models for pit stop decision-making.

Methods and models.

- The race is modeled as a discrete-event system, with events including lap completions, pit stops, overtakes, and incidents.
- Probabilistic models are used to represent lap times and event durations.
- Pit stop strategies are parameterized in terms of the number and timing of stops and evaluated through repeated simulation runs.
- Alternative strategies are compared using statistical measures of total race time and finishing position.

Results.

- Discrete-event simulation can reveal substantial differences between alternative pit stop strategies.
- Stochastic events and traffic have a significant impact on the effectiveness of predefined strategies.
- Optimal strategies depend not only on average race pace but also on variability in race dynamics.
- Simulation-based models are well suited for supporting strategic decision-making in Formula 1.

3 Data Sources

1) Kaggle: Formula 1 World Championship (1950–2024)

https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

The primary data source for this project is the public Kaggle dataset "Formula 1 World Championship (1950–2024)" by rohanrao, which contains historical Formula 1 data including seasons, races, drivers, constructors, lap times, and pit stop events. From this dataset, tables required to link races and participants were extracted, such as races, drivers, and constructors, as well as race events at the lap and pit stop level (lap times, pit stops).

Temporal coverage: the analysis focuses on the 2011–2024 seasons, since pit stop data relevant to the scope of this study is consistently available starting from 2011. This constraint was introduced to ensure data completeness and comparability of strategic events.

2) FastF1: Tire Compounds and Safety Car Data

To incorporate strategy-critical factors such as tire compounds and race neutralization periods, data from the FastF1 library was used. FastF1 provides structured access to Formula 1 timing, telemetry, and session data.

Coverage: 2018–2024, as reliable and consistent information on tire usage and safety car events is available for this period within the implemented data pipeline.

The following variables were obtained from FastF1:

- `tyre_compound` - tire compound used during a given stint,
- `safety_car` - indicator of Safety Car or race neutralization influence.

3) Open-Meteo: Weather Data

Weather conditions were collected using the Open-Meteo API, based on the date and location of each race. Weather information was included to account for external environmental factors that may influence race pace, tire degradation, and strategic decisions.

The following weather variables were collected: temperature, humidity, precipitation, wind speed, type of weather (for example, clear or rain).

4) Kaggle: Formula 1 Race Events

https://www.kaggle.com/datasets/jtrotman/formula-1-race-events

An additional Kaggle dataset by jtrotman was used to track Safety Car deployments and red flag events throughout races. This dataset provided lap-level information on race neutralization periods, which was used to construct the `safety_car` indicator variable for seasons not covered by the FastF1 pipeline.

Feature Set Description

The resulting dataset contains the following groups of variables:

A) Season and Event Metadata

- `year` - championship season year
- `date`, `time` - race date and time
- `round` - round number within the season
- `circuitRef` - race location or Grand Prix

B) Identifiers and Entities

- `driver_code` - abbreviated driver code (for example, VER, HAM)
- `constructor` - team or constructor
- `raceId` - unique race identifier

C) Lap-Level Timing and Pit Stop Data

- `lap` - lap number
- `time_lap` - lap time converted to seconds
- `pit` - binary indicator of a pit stop on the given lap
- `time_pit` - pit stop duration in seconds

D) Stint and Tire Degradation Proxies

- `stint` - stint index between pit stops
- `stint_time_delta` - difference between first lap in stint and curent lap
- `tyre_life` - number of laps completed on the current tire set
- `tyre_compound` - tire compound used during the stint

E) Positional Dynamics

- `position` - driver position on the current lap

F) Race Events

- `safety_car` - indicator of Safety Car or race neutralization periods affecting pace and pit stop value
- `red_flag` - indicator of a red flag event

G) Weather Conditions.

- `temperature` - ambient temperature
- `humidity` - relative humidity
- `precipitation` - precipitation level
- `wind_speed` - wind speed
- `weather_type` - aggregated weather category (dry or rainy)

Data Coverage and Limitations

- Pit stop analysis is conducted for the 2011-2024 period, reflecting the selected working window of the Kaggle dataset.
- Tire compound features are only available for 2018-2024. For earlier seasons, these variables are either missing or require exclusion from models that depend on them.
- Weather data from Open-Meteo is linked to race date and location and can be treated either as race-level conditions or further refined temporally depending on the modeling approach.

4 Data Analysis

4.1 Duration of pit stops

To better understand the structural properties of pit stop strategies and their dependence on circuit characteristics, an exploratory analysis of pit stop frequency across different tracks was conducted. The objective of this analysis is to investigate how the average number and variability of pit stops differ between circuits, and to identify systematic patterns that may influence strategic decision-making.

By examining both the mean pit stop count and the distribution of pit stops per race for each circuit, this analysis provides insights into the extent to which pit stop behavior is driven by track-specific factors as opposed to race-level randomness. Such insights are essential for motivating the inclusion of circuit-related features in subsequent modeling and for understanding the baseline strategic landscape across Formula 1 circuits.

Average Number of Pit Stops per Race by Circuit

The figure 4.1 presents the average number of pit stops per race for each circuit, aggregated over the analyzed seasons. A clear heterogeneity across circuits can be observed, indicating that pit stop frequency is strongly track-dependent.

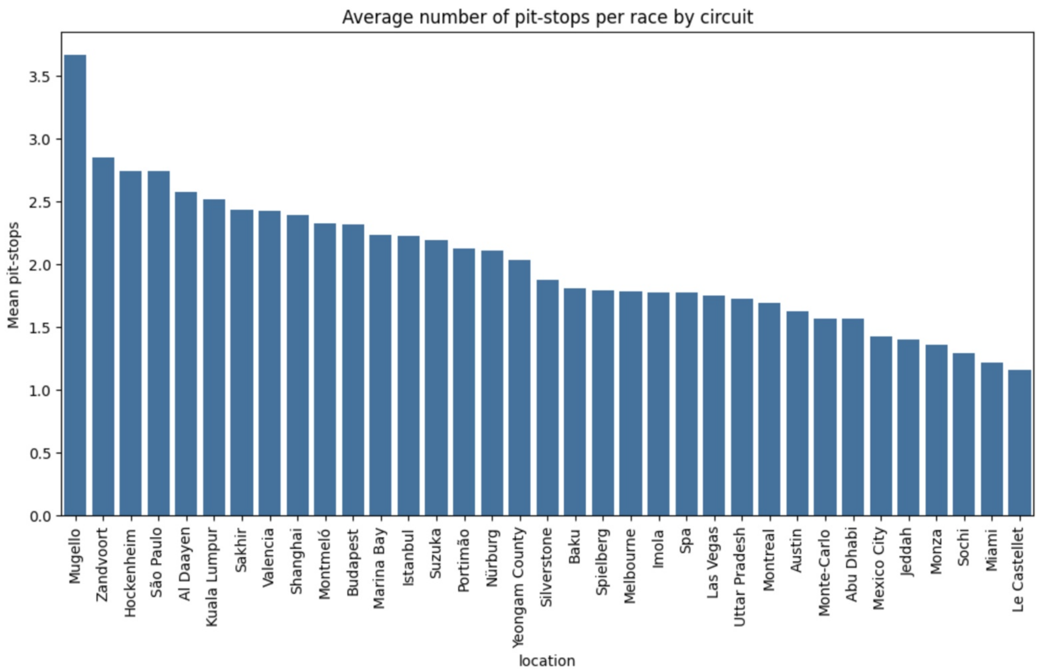

Figure 4.1: Bar plot: Average number of pit-stops by circuit

Circuits such as Mugello, Zandvoort, and Hockenheim exhibit the highest average number of pit stops per race, suggesting higher tire degradation rates or strategic incentives for multiple stops. These tracks are known for layouts that place greater stress on tires or offer limited overtaking opportunities, increasing the strategic value of pit stop timing.

In contrast, circuits like Le Castellet, Miami, Sochi, and Monza show substantially lower average pit stop counts. This pattern is consistent with tracks characterized by smoother asphalt, lower degradation, or race dynamics that favor longer stints and fewer strategic interventions. Overall, the results demonstrate that circuit-specific characteristics play a significant role in determining the baseline pit stop strategy, motivating the inclusion of circuit-level effects in strategic modeling.

Distribution of Pit Stops per Race by Circuit

The figure 4.2 shows the distribution of the number of pit stops per race for each circuit using box plots. This visualization highlights not only differences in central tendency but also the variability and spread of pit stop strategies across races.

Several circuits display relatively tight distributions centered around one or two pit stops, indicating stable and predictable strategic patterns. Other tracks, such as Mugello, Abu Dhabi, and Al Dayeen, exhibit wider distributions and higher upper tails, reflecting greater strategic uncertainty and sensitivity to race-specific factors such as safety car periods, weather conditions, or traffic effects.

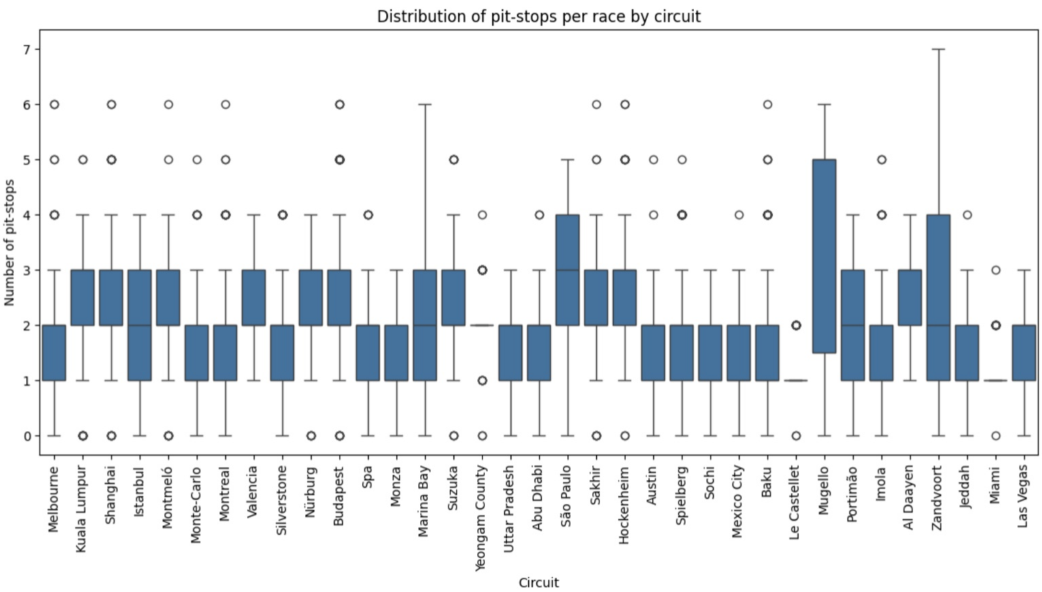

Figure 4.2: Box plot: Distribution of pit-stops per race by circuit

The presence of outliers across many circuits suggests that atypical race events can significantly alter optimal pit stop behavior, leading to strategies with a larger number of stops than usual. This observation supports the need for dynamic and adaptive strategy models rather than fixed heuristic rules.

Taken together, the distributional analysis confirms that pit stop strategies are both circuit-dependent and highly variable, reinforcing the importance of modeling pit stop decisions as a context-aware and data-driven process.

4.2 Analysis of Pit Stop Durations

This section focuses on the analysis of pit stop durations as a critical component of race strategy. Pit stop time directly affects total race time and therefore represents a key cost associated with strategic decisions. Understanding how pit stop durations vary over time and across teams is essential for accurately modeling the trade-offs involved in pit stop strategy optimization.

The following analyses examine temporal trends in pit stop durations across seasons and structural differences in pit stop performance between teams.

Pit Stop Duration by Season

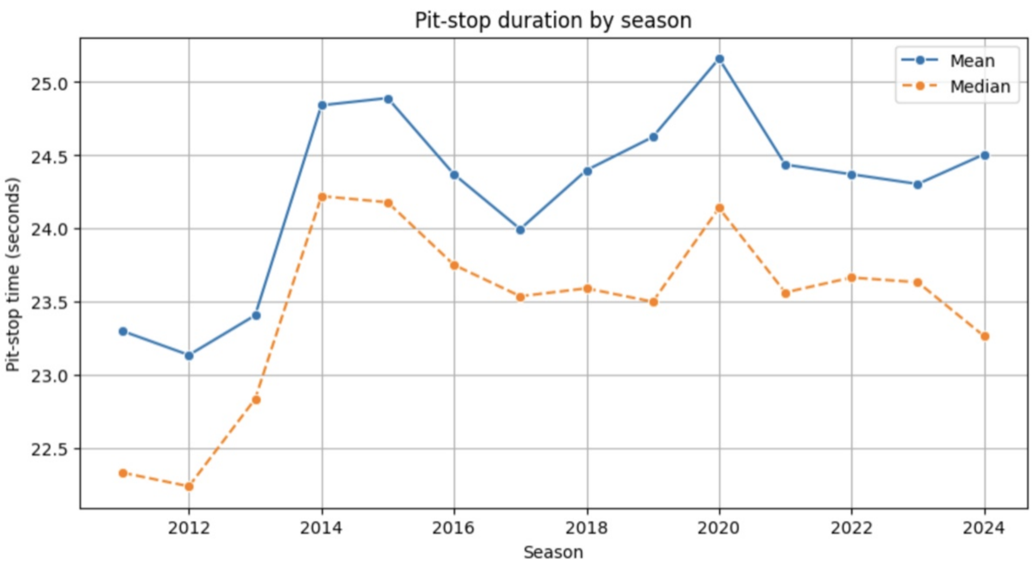

Figure 4.3: Line plot: Pit-stop duration by season

Figure 4.3 illustrates the evolution of pit stop durations across seasons, showing both the mean and median pit stop time for each year. This comparison allows for the assessment of long-term trends as well as the influence of extreme values on average pit stop duration.

Overall, pit stop times remain relatively stable over the analyzed period, with moderate fluctuations between seasons. Differences between mean and median values indicate the presence of outliers, such as unusually slow pit stops, which increase the average duration. The observed variation suggests that while incremental efficiency improvements occur, pit stop execution time remains a persistent and non-negligible strategic cost.

Distribution of Pit Stop Durations by Team

The figure 4.4 presents the distribution of pit stop durations for each team using box plots. This visualization highlights both central tendencies and variability in pit stop performance across teams.

Substantial differences between teams can be observed in both median pit stop time and dispersion. Some teams exhibit consistently faster and more stable pit stop performance, while others show wider distributions and a higher frequency of slow outliers. These results indicate that pit stop execution is not only influenced by season-level factors but also strongly depends on team-specific operational efficiency.

The observed heterogeneity across teams supports the inclusion of team-level features in pit stop strategy modeling and emphasizes the importance of accounting for operational performance when evaluating alternative strategic decisions.

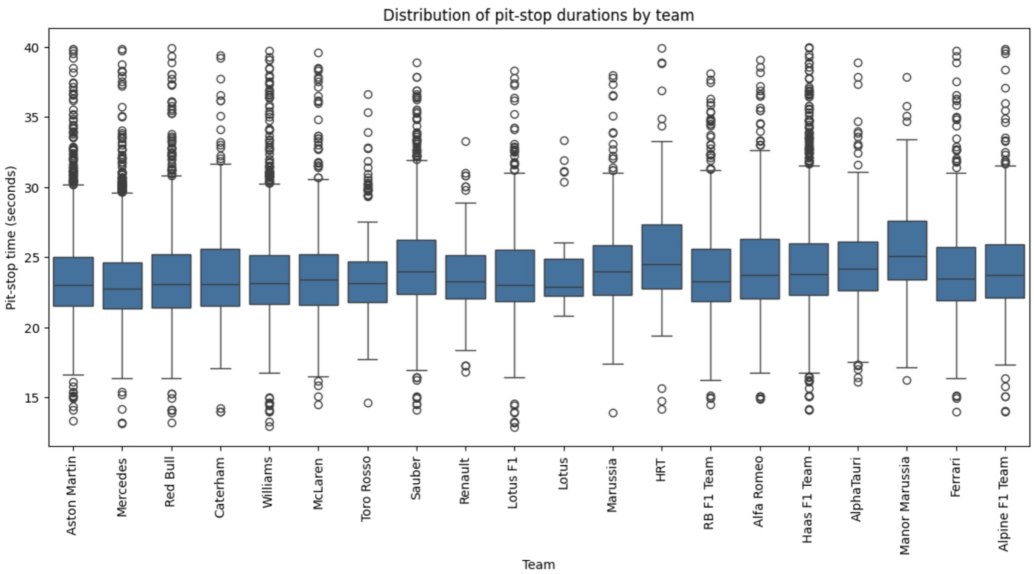

Figure 4.4: Box plot: Distribution of pit-stop duration by constructor

4.3 Analysis of Position Change Around Pit Stops

This section examines the relationship between pit stop timing and subsequent changes in track position. Since pit stops involve a trade-off between immediate time loss and potential performance gains from fresher tires, their timing can have a direct impact on position outcomes. The objective of this analysis is to explore how the lap on which a pit stop is performed, as well as the broader phase of the race, relates to position gains or losses.

The analysis focuses on post-pit position change as an outcome variable, providing insight into the strategic consequences of pit stop timing decisions.

Position Change vs Pit Lap

Figure 4.5 shows the relationship between pit stop lap and the resulting position change. Each point represents a pit stop event, with positive values indicating position gains and negative values indicating position losses. The horizontal reference line at zero highlights the boundary between net position gain and loss.

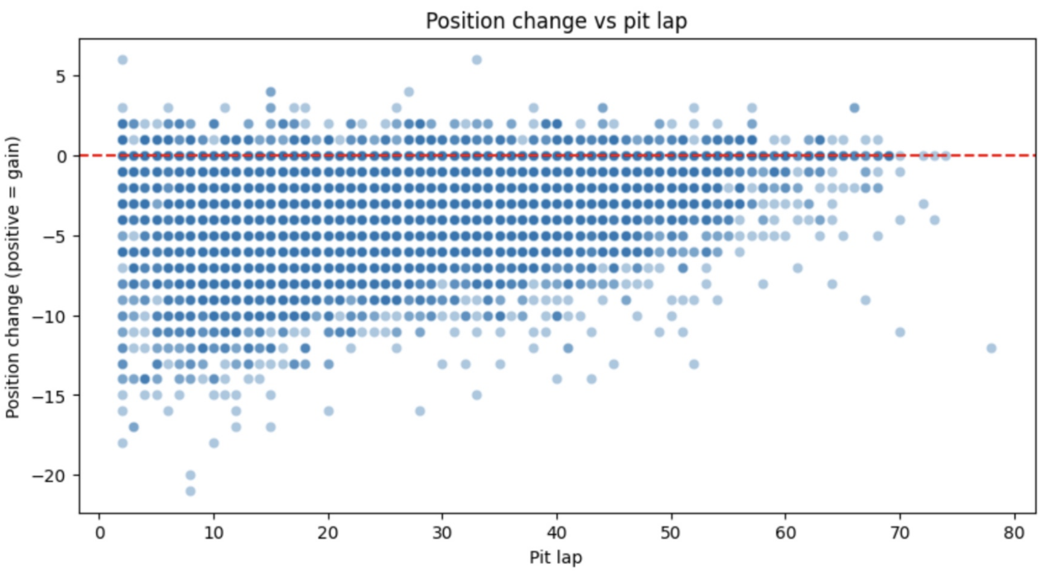

Figure 4.5: Scatter plot: Connection between changing position and pit lap

The scatter reveals substantial variability across pit laps, with most pit stops resulting in a net loss of positions immediately after the stop. This pattern reflects the inherent cost of pit lane time. However, the dispersion of outcomes indicates that pit stops performed at certain laps can mitigate or partially offset this loss, depending on race context such as traffic conditions and competitor strategies. No single pit lap consistently guarantees position gains, underscoring the context-dependent nature of optimal pit timing.

Position Change by Pit Timing Phase

The figure 4.6 summarizes position changes by categorizing pit stops into early, mid, and late race phases. Box plots are used to compare the distributions of position change across these timing categories.

Early pit stops are associated with the largest median position losses and the highest variability, indicating increased strategic risk. Mid-race pit stops show a slightly improved median outcome, while late pit stops are characterized by smaller median losses and a tighter distribution. This trend suggests that pit stops executed later in the race tend to reduce positional risk, potentially due to clearer track conditions or fewer remaining competitors.

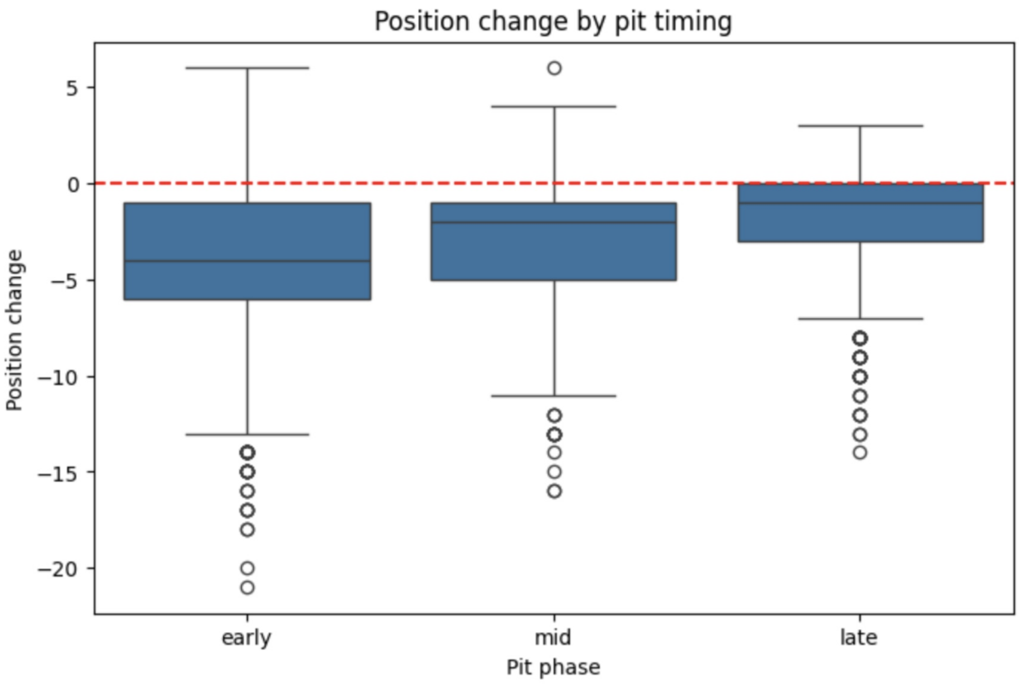

Figure 4.6: Box plot: Position change by pit timing

Overall, the results indicate that pit stop timing plays a significant role in position outcomes and should be treated as a key decision variable in pit stop strategy optimization models.

4.4 Overtaking Window and Strategic Implications

The heatmap (figure 4.7) of the average position change by pit lap percentage highlights the existence of a clear overtaking window during the race. The results show that pit stops performed in the early phase of the race, particularly between 10-20% of race distance, are associated with the largest average position losses. During this phase, high traffic density and small time gaps between cars significantly reduce the ability to exploit fresh tires for overtaking.

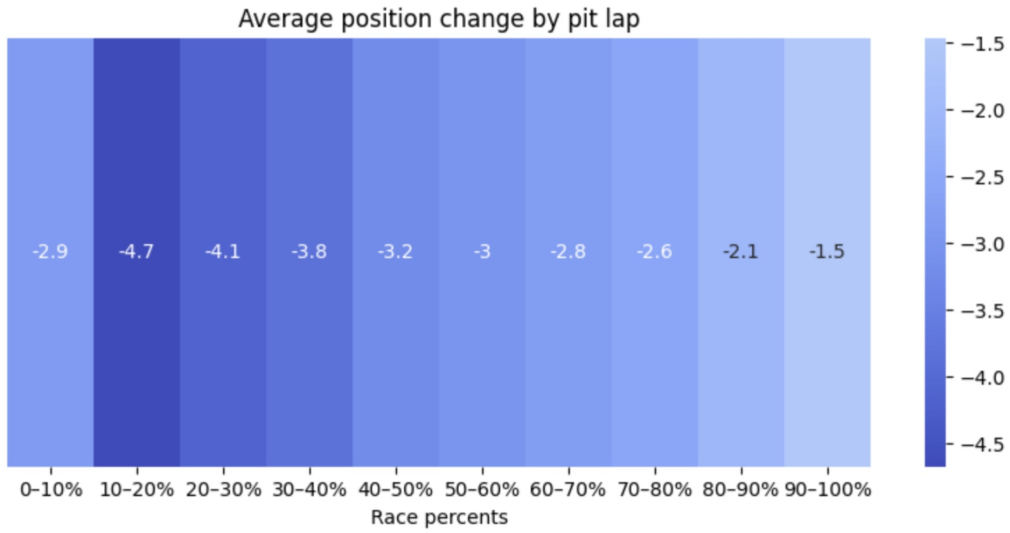

Figure 4.7: Heat Map: Average position change by pit lap

As the race progresses, the average position loss steadily decreases, indicating improved overtaking potential in later phases. The most favorable overtaking window emerges in the final stages of the race, approximately between 80-100% of race distance, where pit stops result in the smallest average position loss. This suggests that reduced traffic and larger time gaps allow drivers to convert tire performance advantages into position gains more effectively.

These findings indicate that overtaking opportunities are not uniformly distributed over the race distance and that pit stop timing relative to this overtaking window is a critical factor in strategy optimization. Incorporating race progress as a normalized feature enables models to capture the interaction between pit stop timing, traffic conditions, and overtaking potential.

4.5 Analysis of Tire Degradation and Stint Structure

Tire degradation is one of the key factors determining race pace evolution and pit stop strategy decisions. As tire performance deteriorates over time, lap times gradually increase, creating a trade-off between continuing a stint and incurring the time cost of a pit stop. Understanding typical stint lengths and the progression of performance loss within a stint is therefore essential for modeling optimal pit stop timing.

This section analyzes the empirical distribution of stint lengths and the average degradation pattern within a stint, providing insight into how long tires are typically used and how performance declines as a function of laps since the last pit stop.

Distribution of Stint Lengths

Figure 4.8 shows the distribution of stint lengths measured in laps. The histogram reveals a right-skewed distribution, with the majority of stints concentrated between approximately 10 and 25 laps. This range corresponds to the most common strategic window in which teams balance tire degradation against pit stop costs.

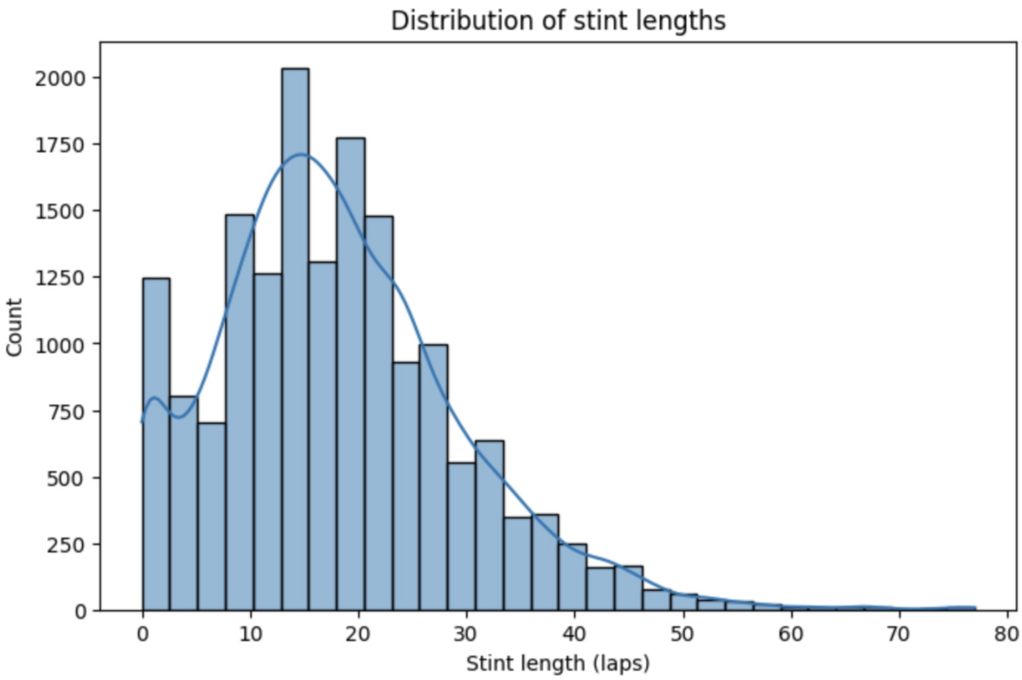

Figure 4.8: Histogram + KDE plot: distribution of stint lengths

Short stints occur less frequently and are often associated with race interruptions, such as safety car periods or tactical deviations. Very long stints, extending beyond 40 laps, are rare and form the tail of the distribution, indicating exceptional circumstances such as conservative strategies or races with limited degradation.

Overall, the distribution suggests that stint length is constrained by degradation dynamics rather than purely strategic preference, reinforcing the importance of modeling tire wear when evaluating pit stop strategies.

Tire Degradation Within a Stint

Figure 4.9 illustrates the average lap time loss relative to the best lap within a stint as a function of the number of laps since the last pit stop. This curve represents the typical degradation profile observed across stints.

The results show a clear monotonic increase in lap time loss as the stint progresses. Performance degradation is relatively rapid in the early laps, followed by a more gradual increase that approaches a plateau after approximately 12 to 15 laps. By the later stages of a stint, the average lap time loss exceeds 0.8 seconds compared to the best lap.

This pattern indicates that tire degradation has a nonlinear structure, with diminishing marginal degradation in the later laps of a stint. Such behavior supports the use of nonlinear or state-based degradation models rather than simple linear approximations when simulating race pace and optimizing pit stop timing.

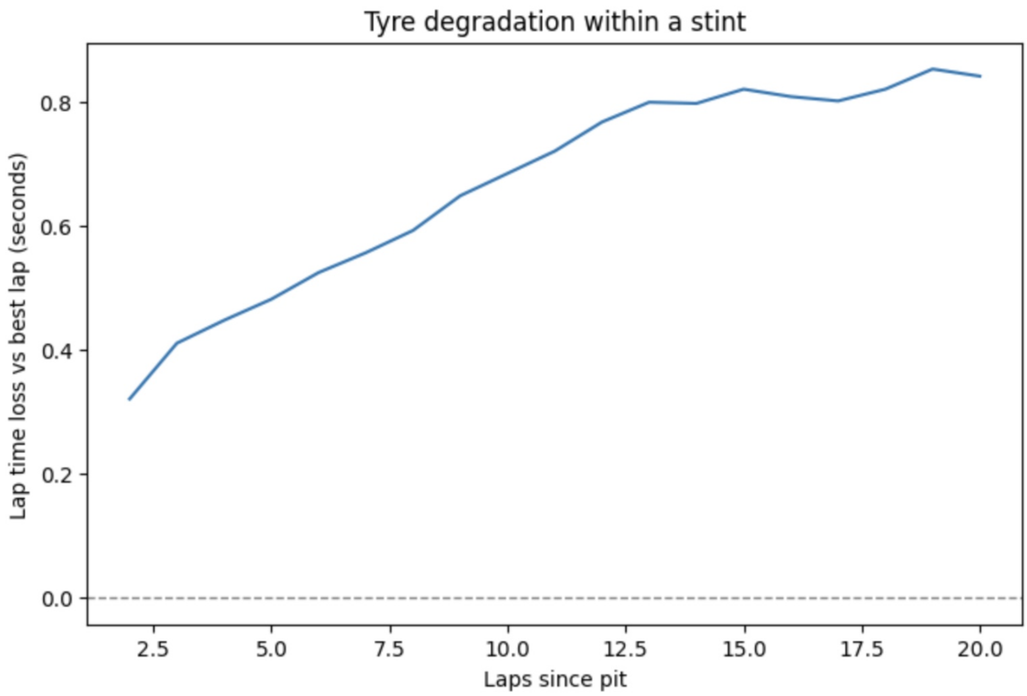

Figure 4.9: Line Plot: Tyre degradation within a stint

4.6 Impact of Safety Car Periods on Pit Stop Outcomes

Safety Car periods represent one of the most influential external factors affecting pit stop strategy. During Safety Car conditions, the field is compressed and lap times are significantly reduced, which lowers the effective time loss of a pit stop compared to green-flag conditions. As a result, pit stops executed under Safety Car can substantially alter the risk-reward balance of strategic decisions.

This section analyzes how the presence of a Safety Car at the time of a pit stop affects subsequent position changes, providing empirical evidence of the strategic value of Safety Car timing.

Position Change on Pit Stop: Safety Car vs Green Flag

The figure 4.10 compares position changes following pit stops performed under Safety Car conditions and under normal green-flag racing. Positive values correspond to position gains, while negative values indicate position losses.

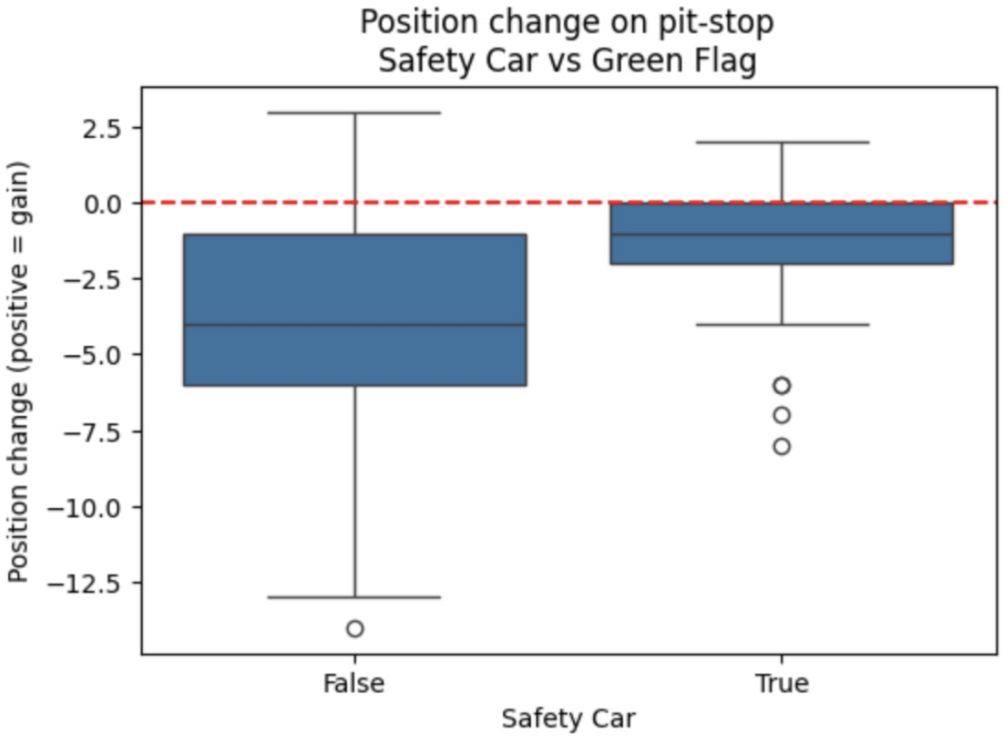

Figure 4.10: Box plot: Position change without safety-car and changing with safety-car

Pit stops executed under green-flag conditions are associated with larger median position losses and a wider distribution, reflecting the full time cost of pit lane entry and rejoining in traffic. In contrast, pit stops performed during Safety Car periods show a noticeably smaller median position loss and reduced variability.

The results indicate that Safety Car periods significantly mitigate the positional cost of a pit stop and, in some cases, allow drivers to preserve or even gain positions. This finding confirms the strategic importance of exploiting Safety Car windows and supports the inclusion of Safety Car indicators as a critical contextual feature in pit stop strategy optimization models.

4.7 Analysis of Tire Compound Effects on Pit Stop Strategy

Tire compound selection is a central component of pit stop strategy, as different compounds exhibit distinct trade-offs between grip, degradation rate, and usable stint length. The choice of tire compound influences both race pace evolution and the strategic value of a pit stop, particularly in terms of position outcomes immediately after the stop. This section examines empirical patterns in tire compound usage and analyzes how pit stop outcomes vary across different tire compounds.

Tire Compound Usage Frequency

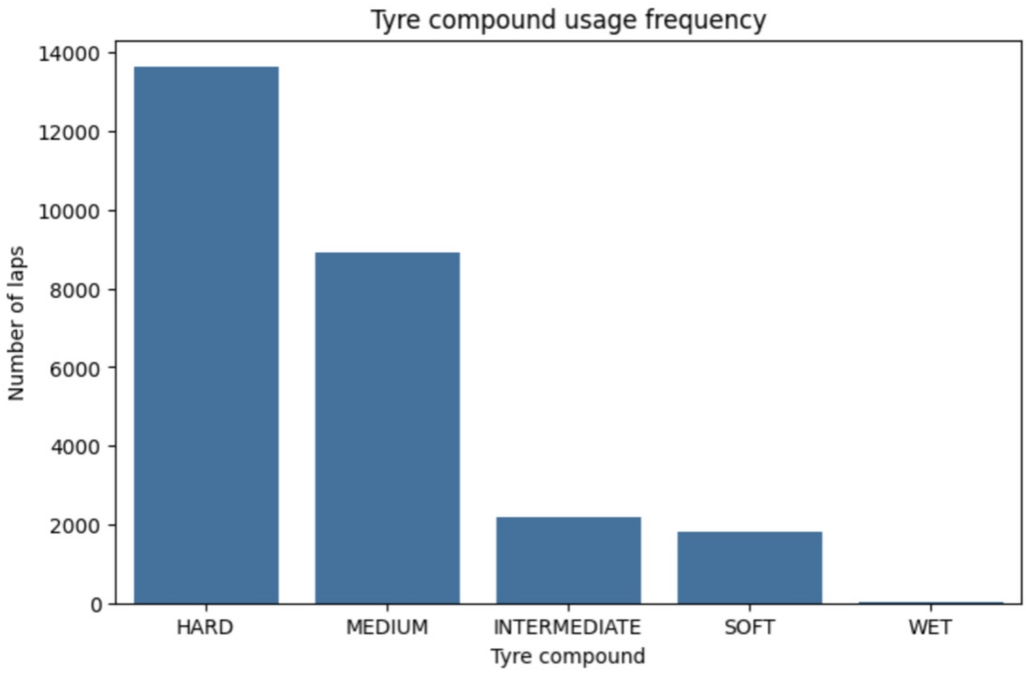

Figure 4.11: Count plot: Tyre compound usage frequency

Figure 4.11 shows the frequency of tire compound usage measured by the total number of laps completed on each compound. Hard and medium compounds dominate overall usage, reflecting their durability and suitability for longer stints under typical race conditions.

Intermediate and wet tires are used substantially less frequently, as their deployment is restricted to specific weather conditions. Soft tires also account for a smaller share of total laps, indicating that their higher degradation rates limit stint length despite their short-term performance advantage.

This distribution highlights that most strategic decisions are made under dry conditions using hard and medium compounds, motivating a primary focus on these compounds in baseline strategy modeling.

5 Research Hypotheses

H1. Pit Timing Hypothesis. The relative timing of a pit stop within the race significantly affects post-pit position change. Pit stops performed later in the race are associated with smaller average position losses compared to early-race pit stops.

H2. Safety Car Mitigation Hypothesis. Safety Car presence significantly reduces the positional cost of a pit stop. Pit stops executed under Safety Car conditions result in smaller median position losses compared to green-flag pit stops.

H3. Tire Degradation Structure Hypothesis. Lap time degradation within a stint follows a nonlinear increasing pattern. Lap time loss increases as a convex function of laps since pit, with diminishing marginal increase in later laps.

H4. Team Operational Effect Hypothesis. Team-level operational efficiency significantly influences pit stop outcomes. Teams with lower median pit stop durations experience smaller average position losses.

H5. Circuit Strategy Hypothesis. The optimal number of pit stops varies significantly across circuits. Circuits can be categorised by their historical median pit stop count, and strategies matching the circuit-specific category produce lower total race times compared to strategies with a fixed number of stops applied uniformly across circuits.

Testing Hypotheses

H1. Pit Timing Hypothesis

The hypothesis states that later pit stops are associated with smaller position losses compared to early pit stops. To test this, all pit stops were divided into three phases by lap fraction: early (`lap_frac` < 0.33), mid (0.33–0.66), and late (> 0.66). The target variable was the change in position on the pit stop lap relative to the previous lap. The Kruskal-Wallis test was applied to compare distributions across groups, followed by pairwise Mann-Whitney tests.

The Kruskal-Wallis test revealed statistically significant differences between groups ($H = 89.3$, $p < 0.001$). Mean position losses were 1.06 for the early phase, 0.87 for the mid phase, and 0.55 for the late phase - a clear decreasing trend. Pairwise Mann-Whitney tests confirmed that late pit stops are statistically significantly better than both early and mid pit stops ($p < 0.001$ in both cases). The difference between early and mid phases was not significant ($p = 0.16$).

Hypothesis H1 is confirmed – later pit stops are associated with smaller position losses, and the difference is statistically significant.

H2. Safety Car Mitigation Hypothesis

The hypothesis states that pit stops executed under Safety Car conditions result in smaller position losses compared to green-flag pit stops. To test this, pit stops were divided into two groups based on Safety Car status on the pit lap. A one-sided Mann-Whitney test was applied.

Mean position losses were 0.22 under SC compared to 0.92 under green flag conditions — a difference of more than four times. The Mann-Whitney test confirmed that the distribution of position losses under SC is statistically significantly smaller than under green flag conditions ($p < 0.001$).

Hypothesis H2 is confirmed - Safety Car presence significantly reduces the positional cost of a pit stop.

H3. Tire Degradation Structure Hypothesis

The hypothesis states that lap time degradation within a stint follows a nonlinear increasing pattern with diminishing marginal increase. To test this, median values of clean degradation (`target_degrad_clean`) by `tyre_life` in the range of 3–35 laps were used. Nonlinearity was tested via an F-test comparing linear and polynomial (degree 2) regressions.

The F-test showed a statistically significant improvement when adding the quadratic term ($F = 330.0$, $p < 0.001$). The polynomial model explains 98.2% of variance compared to 78.2% for the linear model. The negative coefficient on x^2 (-0.002) confirms the concave shape of the curve — the mean degradation increment in the early stint (laps 3–15) is 0.081 s/lap compared to 0.014 s/lap in the later laps (laps 15–35), a sixfold decrease.

Hypothesis H3 is confirmed - degradation is nonlinear and exhibits diminishing marginal increase, however the shape of the curve is concave rather than convex as stated in the hypothesis.

H4. Team Operational Effect Hypothesis

The hypothesis states that teams with lower median pit stop durations experience smaller average position losses. To test this, median pit stop time and mean position change after a pit stop were calculated for each team. Spearman correlation was applied to assess the relationship between these variables - a nonparametric method robust to outliers that does not require normality assumptions.

A negative correlation in the expected direction was observed ($r = -0.42$) - teams with faster pit stops tend to lose fewer positions on average. However, the correlation does not reach statistical significance ($p = 0.066$). A partial explanation is the small number of observations at the team level (20 teams), which reduces statistical power, as well as the fact that position losses are determined not only by pit stop time but also by track position, traffic, and competitor strategic decisions.

Hypothesis H4 is not confirmed - the expected trend is observed but does not reach statistical significance.

H5. Circuit Strategy Hypothesis

The hypothesis states that the optimal number of pit stops varies significantly across circuits and that circuits can be categorised by their historical median pit stop count. To test the first part of the hypothesis, the number of pit stops per driver per race was used across all circuits in the dataset. The Kruskal-Wallis test was applied to compare distributions across circuits and categories, followed by pairwise Mann-Whitney tests.

The Kruskal-Wallis test across 35 circuits revealed highly significant differences ($H = 1156.9$, $p < 0.001$). Circuits were divided into three categories by median pit stop count: 1-stop (8 circuits, median = 1.0), 2-stop (22 circuits, median = 2.0), and 3-stop (5 circuits, median = 3.0). Differences between all three categories are also highly significant ($H = 775.5$, $p < 0.001$), and pairwise Mann-Whitney tests confirmed that each category is statistically significantly different from the adjacent one ($p < 0.001$ in both cases).

Hypothesis H5 is partially confirmed - the number of pit stops varies statistically significantly across circuits and the categorisation based on historical data is valid. The second part of the hypothesis, that strategies matching the circuit-specific category produce lower total race times, is supported by simulation results - grid search using circuit categories found strategies that improve upon the baseline on both test circuits.

6 Simulation enviroment

The central component of this work is a lap-by-lap race simulation environment for Formula 1, developed to evaluate and compare pit stop strategies under stochastic race conditions. The environment reproduces race dynamics at the level of individual laps: at each step, an agent observes the current race state and makes a strategic decision - whether to continue on the current tyres or pit with a specific compound choice.

The need for a custom simulation environment arises from the fact that real telemetry data reflects only the strategies that were actually executed, whereas evaluating alternative decisions requires the ability to replay identical race conditions with different strategic scenarios. The environment supports thousands of simulations under fixed initial conditions - circuit, driver, constructor, weather - varying only the strategic decisions. This enables probabilistic evaluation of strategies through median and mean race time across multiple scenarios with varying Safety Car outcomes.

Lap time in the simulation is computed as the sum of five components. Base pace - predicted by a model from circuit, constructor, driver, and weather features, and held constant throughout the race. Tyre degradation - deviation from base pace due to tyre wear, accumulating over the course of a stint. Track evolution - gradual improvement of track conditions as rubber builds up on the asphalt surface, common to all participants. Safety Car penalty - slowdown during neutralisation periods, dependent on the lap number within the SC period. Pit stop time loss - added on the lap on which tyres are changed.

Each component is modelled separately from historical data spanning the 2011–2024 seasons using machine learning methods. This modular architecture allows individual components to be updated and improved independently, and provides explicit control over the contribution of each factor to total race time.

The environment enforces two hard constraints through action masking: the Pirelli rule, requiring the use of at least two different dry tyre compounds per race, and a maximum stint length constraint based on the 95th percentile of historical data per compound (SOFT - 31 laps, MEDIUM - 38, HARD - 47). These constraints reflect real regulatory requirements and physical tyre wear limits.

Three approaches to pit stop strategy optimisation are implemented and compared within the simulation environment. The rule-based baseline reproduces standard team behaviour - pitting when tyre life reaches the 60th percentile of historical stint lengths for the given circuit and compound. Grid search performs exhaustive enumeration of reasonable one-stop and two-stop strategies via Monte Carlo simulation and identifies the optimal fixed strategy for given conditions. The Lookahead Policy is an online algorithm that makes a decision at each lap by analytically comparing the expected cost of continuing on current tyres against an immediate pit stop over a 20-lap horizon, with real-time adaptation to Safety Car periods.

6.1 Base Pace Model

Base pace is the reference point of the simulation, relative to which all other lap time components are computed: tyre degradation, track evolution, and SC penalty. It is defined as the median lap time over laps 3-6 of the first stint for a specific driver in a specific race - this range is chosen because tyres are already warmed up, fuel load is still high and stable, and the probability of SC or traffic is minimal.

Since base pace is unknown before the race starts, it must be predicted in advance for simulation purposes. The model predicts base pace from the following features: circuit identifier, constructor, driver, weather conditions (temperature, humidity, precipitation, wind speed, weather type), year, and starting position - the driver's position at the end of lap 1, which reflects qualifying pace. The most important feature by a large margin is circuit (importance=0.45), which is expected since lap times vary fundamentally across circuits: from 70 seconds at Monaco to 95 seconds in Singapore.

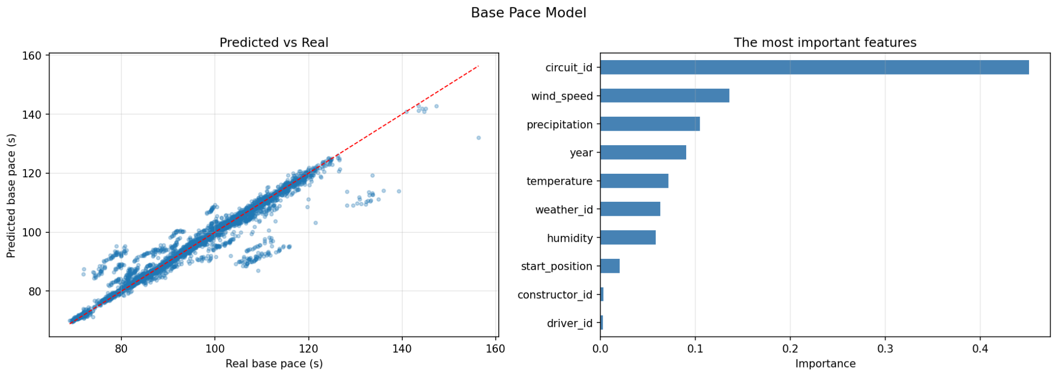

Figure 6.1: Base Pace Model - Predicted vs Real and Feature Importance

The model is implemented using XGBoost with GroupShuffleSplit by raceId to prevent data leakage. Performance: MAE=5.3s, R^2=0.64. An MAE of 5 seconds relative to absolute lap times of 70-150 seconds is acceptable - the primary source of error is that a constructor and driver in a specific year may differ substantially from their historical average, and this information is not available in advance. In simulation, base pace is predicted once at race initialisation and remains constant throughout the race - it is assumed that the car maintains a stable pace without accounting for traffic or positional battles.

6.2 Track Evolution model

As a race progresses, lap times gradually improve for all cars regardless of the tyre condition of any individual driver. This phenomenon - known as track evolution or the rubber-in effect - occurs as tyres deposit rubber onto the asphalt surface, increasing grip levels and reducing lap times for all participants simultaneously. Track evolution is a systematic component of race dynamics that, if left unaccounted for, masks tyre degradation: without correcting for this effect, tyres appear to get faster over the course of a stint, which is physically incorrect.

To estimate the track evolution signal, laps with fresh tyres were selected (tyre life 4-8 within a stint), where tyre wear is minimal and any systematic trend in lap time reflects changes in track conditions rather than tyre behaviour. Lap times were normalised relative to the race median to remove circuit- and driver-specific differences. The resulting curve, averaged across all races in the dataset, shows a monotonic decrease in lap time from the start to the end of the race: at race start the track is approximately +3.3 seconds slower than base pace, while by the end it is -1.7 seconds faster.

Three functional forms were evaluated to approximate this curve: linear, second-degree polynomial, and logarithmic (figure 6.2).

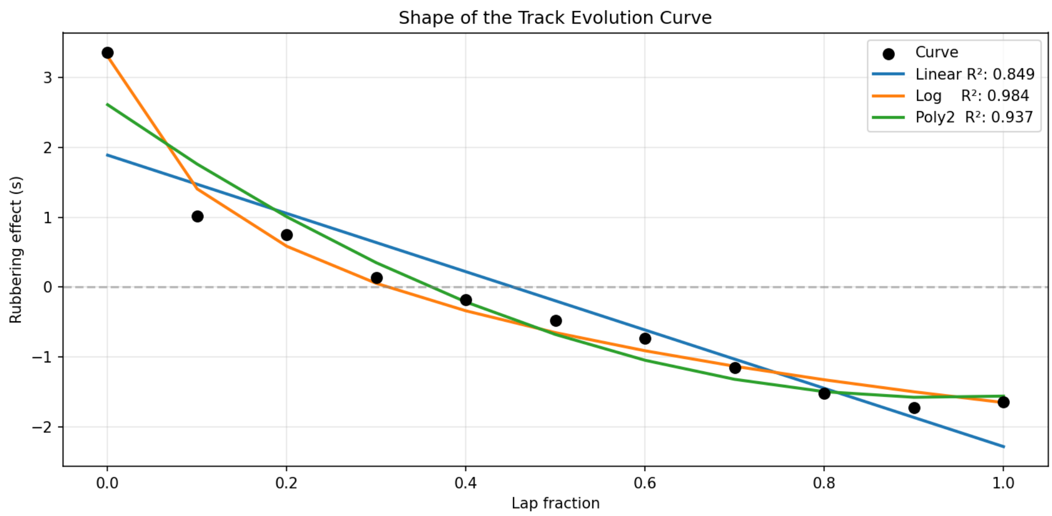

Figure 6.2: Shape of the Track Evolution Curve - fitted functions vs empirical data

The logarithmic specification provided the best fit (R^2=0.984) and is physically interpretable - track evolution is most rapid in the early laps and gradually plateaus as the surface reaches rubber saturation.

The estimated function is applied in the simulation as a correction term subtracted from lap time at each step, anchored to laps 3-6 of the first stint as the reference point consistent with the base pace definition used in tyre degradation modelling.

The importance of this correction is illustrated in the figure below. Without subtracting track evolution, the degradation signal for all compounds becomes negative - tyres appear to get faster with each lap, which is physically meaningless. After applying the correction, the signal becomes positive and increases with tyre life, as expected.

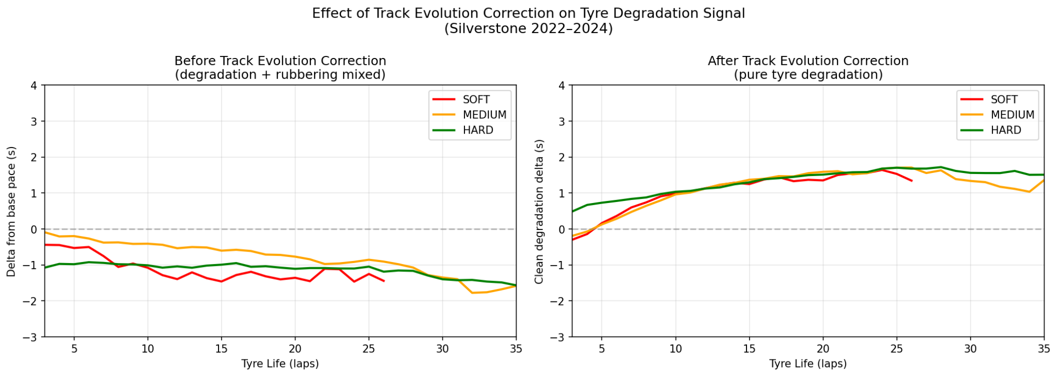

Figure 6.3: Effect of Track Evolution Correction on Tyre Degradation Signal - before and after correction

6.3 Tyre Degradation Model

Tyre degradation is one of the key factors determining pit stop timing. As a stint progresses, tyre performance deteriorates and lap times increase, creating a trade-off between continuing on worn tyres and incurring the time cost of a pit stop. Accurately modelling this process requires isolating the pure degradation signal from other sources of lap time variation - most importantly track evolution and the fuel load effect.

Target Construction

The degradation target is constructed in three steps. First, a driver's base pace is defined as the median lap time over laps 3–6 of the first stint, absorbing circuit, constructor, and driver effects. Second, the fuel effect is accounted for: as the race progresses, the car becomes lighter as fuel is consumed (∼1.8 kg per lap), which in itself accelerates the car by approximately 0.03 seconds per kilogram. This effect is computed analytically for each lap and subtracted from the target. Finally, track evolution is subtracted from the lap time delta relative to base pace, yielding a clean degradation signal:

$$\texttt{target} = \texttt{time\_lap} - \texttt{base\_pace} - \texttt{fuel\_effect} - \texttt{rubbering\_corrected} \tag{1}$$

This target reflects only the lap time cost attributable to tyre wear - it is anchored to zero at the start of a stint and increases as tyres age.

Features

The following feature set is constructed for each observation:

- **Tyre life and its transformations** - `tyre_life`, $\texttt{tyre\_life}^2$, $\texttt{tyre\_life}^3$, $\sqrt{\texttt{tyre\_life}}$, $\log(1 + \texttt{tyre\_life})$. Polynomial and nonlinear transformations allow the model to capture the nonlinear degradation profile - rapid warm-up in the early laps followed by gradual wear accumulation.

- **Compound-wear interaction** - $\texttt{compound\_num} \times \texttt{tyre\_life}$. Different compounds degrade at different rates, and this interaction term allows the model to adapt the degradation curve to the specific tyre type.

- **Lap fraction** - the proportion of the race completed (0 to 1). Captures residual track evolution and changing track conditions not fully absorbed by the correction function.

- **Circuit** - `circuit_id` and `circuit_degrad_tendency` (median degradation per circuit computed from training data only). Circuits differ substantially in surface abrasiveness and thermal regime.

- **Weather** - temperature, humidity, precipitation, wind speed, `weather_id`. High temperatures accelerate tyre wear; precipitation changes the degradation profile entirely.

- **Fuel effect** - `fuel_effect`, computed as $(\texttt{total\_laps} - \texttt{lap}) \times 1.8 \times 0.03$. Passed as a feature so the model can adjust predictions based on the current fuel load of the car.

Models and Results

A separate XGBoost model is trained for each tyre compound (SOFT, MEDIUM, HARD, INTERMEDIATE) on data from 2021–2024 where compound information is available, plus a general model covering the full 2011–2024 period. Training uses an 80/20 random split.

MAE of approximately 0.5 seconds per lap is consistent with the inherent noise in lap time data arising from traffic, minor incidents, and driver variability. The relatively modest R^2 values reflect the stochastic nature of the target rather than model underfitting - the residual variance is largely irreducible given the available features.

To prevent unrealistic extrapolation beyond historically observed stint lengths, an exponential penalty is applied when `tyre_life` exceeds the 95th percentile of historical stint lengths for each compound (SOFT: 31 laps, MEDIUM: 38, HARD: 47). This ensures the model does not produce optimistic degradation estimates in a regime where no training data exists.

Table 6.1: Tyre degradation model performance by compound.

Model	**MAE**	**R^2**	**Train Sample**
All	0.672	0.447	174 107
Soft	0.502	0.500	5 735
Medium	0.512	0.586	17 108
Hard	0.531	0.522	24 556
Intermediate	0.715	0.715	2 106

6.4 Safety Car Model

Safety Car is one of the key stochastic events in a race, with a significant impact on strategic decisions. During a neutralisation period, the pace of the entire field equalises, which reduces the effective cost of a pit stop and creates an opportunity for a free tyre change. Modelling the probability and duration of SC periods is therefore essential for accurate strategy evaluation in simulation.

A two-level probabilistic model is implemented. The first level - the race-level model - predicts the probability of at least one SC occurring during a race, based on circuit characteristics, weather conditions, and season. The second level - the lap-level model - predicts the probability of an SC period starting on a specific lap. Both models are implemented using XGBoost. To prevent data leakage, training is performed with GroupShuffleSplit by `raceId` - laps from the same race are never split across training and validation sets. The historical SC rate per circuit (`circuit_sc_rate`) is computed using training races only, to avoid leakage of information about test circuits.

The lap-level model features include lap fraction, weather conditions, circuit identifier, year, and historical SC rate per circuit. The latter proved to be the most important feature - the model relies heavily on the historical prior per circuit, which reflects reality: the probability of SC at Monaco or Baku is objectively higher than at Monza.

Lap-level model performance is ROC-AUC = 0.62, race-level ROC-AUC = 0.48. The modest values reflect the fundamental unpredictability of Safety Car - it is a stochastic event arising from random on-track incidents, and its predictability is limited by the nature of the phenomenon itself rather than by model quality.

SC period duration is sampled randomly from the empirical distribution of historical SC periods - 195 periods are recorded in the data with a mean duration of 4.9 laps and std = 2.1. This approach preserves realistic variability: some SC periods last 2–3 laps while others extend to 8–10 laps.

The lap time penalty under SC is also modelled from data, with an important nuance: the penalty is not constant throughout the SC period. On the first SC lap, drivers do not yet know exactly where the safety car is, brake sharply, and queue up - this is the slowest lap, with time losses of approximately +33–41 seconds relative to normal race pace. On subsequent laps the field is already organised and travelling steadily behind the SC - the penalty gradually decreases and stabilises at around +30 seconds by laps 8–10. These values are estimated as the median difference between lap times under SC and normal pace within the same race, averaged by lap number within the SC period.

6.5 Pit Stop Duration Model

Pit stop time represents a fixed time loss at each tyre change, directly affecting total race time. Unlike tyre degradation which accumulates gradually, a pit stop is a one-off large loss (∼23–25 seconds) that must be offset by the benefit of fresh tyres. Accurate modelling of pit stop duration is important for correctly evaluating the trade-off between early and late pit stop timing.

The model predicts pit stop time from four groups of features. Circuit (`circuit_id`) is the dominant feature (importance = 0.46) - pit lane length varies substantially between circuits and determines the minimum possible pit stop time. Constructor (`constructor_id`) reflects the operational efficiency of the mechanics - top teams consistently execute pit stops faster. Weather (`weather_id`) influences pit stop time through tyre type selection and working conditions. Safety Car status (`safety_car`) - under SC teams operate under different conditions, which may affect service speed.

The model is implemented using XGBoost with GroupShuffleSplit by `raceId` to prevent data leakage. Performance: MAE = 1.7 s, $R^2 = 0.42$.

6.6 Using the Simulation Environment

The environment takes as input the parameters of a specific race - circuit, constructor, driver, weather conditions, and starting position - and reproduces the race lap by lap, generating a complete event history: lap times, Safety Car periods and their durations, pit stops, and total race time. Running the same configuration multiple times with different random seeds produces a distribution of possible outcomes for a given strategy - this allows strategies to be evaluated not against a single deterministic scenario but in a probabilistic sense, explicitly accounting for Safety Car uncertainty and other stochastic race events. In addition to offline optimisation through enumeration of candidate strategies, the environment supports an online mode - making decisions in real time during a simulated race, adapting to the current state including Safety Car deployment and actual tyre degradation within the current stint.

7 Strategies

7.1 Baseline Strategy

As a reference point for comparing optimisation algorithms, a rule-based baseline strategy is implemented that reproduces typical team behaviour. The logic is straightforward: pit when `tyre_life` reaches the 60th percentile of historical stint lengths for the given circuit and compound. The threshold is computed separately for each circuit-compound combination from training data. The tyre sequence is fixed: SOFT at the start, followed by MEDIUM, then HARD.

The choice of the 60th percentile corresponds to a position of "slightly later than the median driver" - most teams have already pitted, but the tyres have not yet reached critical wear. This is a reasonable conservative strategy that requires no predictions and serves as a lower bound for evaluating more sophisticated algorithms.

For Silverstone 2024, the baseline produces a 2-stop strategy with pit stops at laps 15 and 38. Over 300 Monte Carlo simulations, the median race time was 81.728 minutes - this is the reference value against which the improvement of all subsequent algorithms is measured.

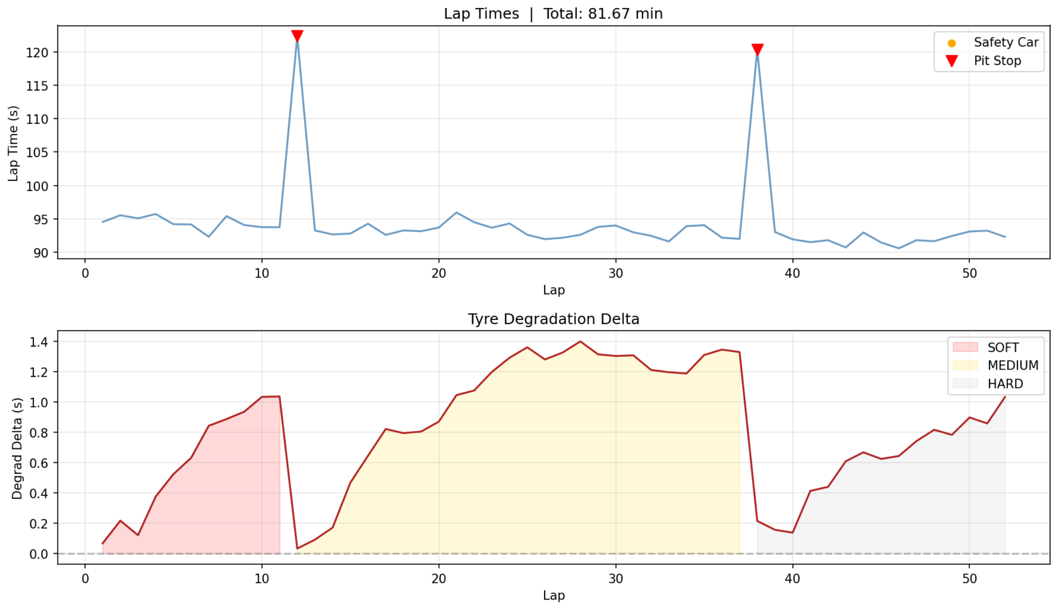

Figure 7.1: Example baseline race simulation - Silverstone 2024, Mercedes, HAM, dry conditions. Top: lap times with pit stop markers. Bottom: tyre degradation delta by compound across the race distance.

7.2 Grid Search

Grid search is an exhaustive enumeration of all reasonable pit stop strategies, with each strategy evaluated through Monte Carlo simulation. Before generating strategies, the circuit is automatically classified by number of pit stops based on historical data - the median number of pit stops per driver per race for each circuit across all seasons in the dataset. Circuits are divided into three categories (table 7.1): 1-stop (Monaco, Monza, Jeddah and others with low degradation - median $\leq$ 1.5), 2-stop (Silverstone, Spa, Bahrain and most others - median 1.5–2.5), 3-stop (Interlagos, Hockenheim, Sepang - median $>$ 2.5). Only strategies corresponding to the relevant category are generated, focusing the search on a physically meaningful space and avoiding clearly suboptimal options.

Table 7.1: Track categories by number of pit stops.

Category	**Circuits**
1-stop (8 circuits)	Monaco, Miami, Jeddah, Imola, Sochi, Rodriguez, Ricard, Monza
2-stop (22 circuits)	Marina Bay, Hungaroring, Catalunya, Buddh, Albert Park, Americas, Bahrain, Baku, Suzuka, Valencia, Silverstone, Istanbul, Red Bull Ring, Portimao, Shanghai, Nurburgring, Vegas, Villeneuve, Spa, Yas Marina, Yeongam, Zandvoort
3-stop (4 circuits)	Losail, Hockenheim, Interlagos, Sepang

For each category, strategies are constructed as all combinations of pit stop laps and tyre compounds within reasonable ranges. For 2-stop strategies this means enumerating all pairs (`pit1`, `pit2`) with a step of 1/2/3 laps and two possible compound sequences - MEDIUM $\rightarrow$ HARD and HARD $\rightarrow$ MEDIUM. Each strategy is evaluated through 50 simulations with different random seeds and scored by median total race time. Median is preferred over mean as it is robust to rare outliers where a Safety Car radically alters the outcome of an individual simulation. After enumeration, strategies are ranked by median and the best is selected.

7.3 Lookahead Policy

Lookahead Policy is an online strategy decision algorithm that operates in real time. Unlike grid search, which finds the optimal strategy in advance through exhaustive enumeration, Lookahead makes a decision at each lap independently - observing the current race state and analytically evaluating two alternative scenarios: continuing on current tyres or pitting immediately.

How It Works

At each lap, two costs are computed over a planning horizon $H = \min(\texttt{laps\_remaining}, 20)$:

$$\texttt{cost\_no\_pit} = \sum_{i=0}^{H} \max\big(0,\ \texttt{degrad}(\texttt{tyre\_life} + i,\ \texttt{compound},\ \texttt{lap\_frac} + i/\texttt{total\_laps})\big)$$
$$+ \texttt{PIT\_TIME\_BASE} \times \left(1 - \frac{\texttt{laps\_to\_max}}{\texttt{MAX\_TYRE\_LIFE}}\right), \quad \text{if } \texttt{laps\_to\_max} \leq \texttt{laps\_rem} \tag{2}$$

$$\texttt{cost\_pit} = \texttt{pit\_time} + \sum_{i=0}^{H} \max\big(0,\ \texttt{degrad}(i,\ \texttt{new\_compound},\ \texttt{lap\_frac} + i/\texttt{total\_laps})\big) \tag{3}$$

If `cost_pit` $<$ `cost_no_pit`, the policy pits and selects the compound with the minimum expected degradation over the horizon. Otherwise it continues.

Safety Car Adaptation

Safety Car fundamentally changes the cost balance. Under normal conditions a pit stop costs ∼23–30 seconds. Under SC the entire field slows down, adding +33–41 seconds to normal pace on each SC lap. As a result, the car on the pit lane loses approximately the same time as all others lose on track - the pit stop becomes effectively free.

In the algorithm this is reflected by replacing `pit_time` with -10.0 under SC - the value captures the net benefit of pitting during a neutralisation period. The policy does not pit unconditionally under SC: it additionally checks that tyres are sufficiently worn (`tyre_life` ≥ 5) and that at least 15 laps remain — otherwise fresh tyres would not recover the pit stop cost.

Constraints

- `tyre_life` < 15 under normal conditions and `tyre_life` < 5 under SC - pit is blocked.
- Pirelli rule: if ≤ 15 laps remain and only one compound has been used - no-pit is blocked.
- Horizon H is capped at 20 laps - degradation uncertainty beyond this exceeds model accuracy (MAE ∼0.5 s).

8 Strategy Comparison

8.1 Race with 2 pit-stops

All three algorithms were tested under identical conditions: Silverstone 2024, Mercedes, Hamilton, dry weather, starting on SOFT. Silverstone is classified as a 2-stop circuit - historically drivers make two pit stops here. 100 Monte Carlo simulations with independent random seeds were run for each strategy.

All three strategies produced similar median race times in the range of 81.67–81.72 minutes, with a difference of only 3–4 seconds between the best and worst result. All distributions are right-skewed - rare simulations with Safety Car produce times of 83–86 minutes, while the majority of simulations are concentrated in the 81.5–82.0 minute range.

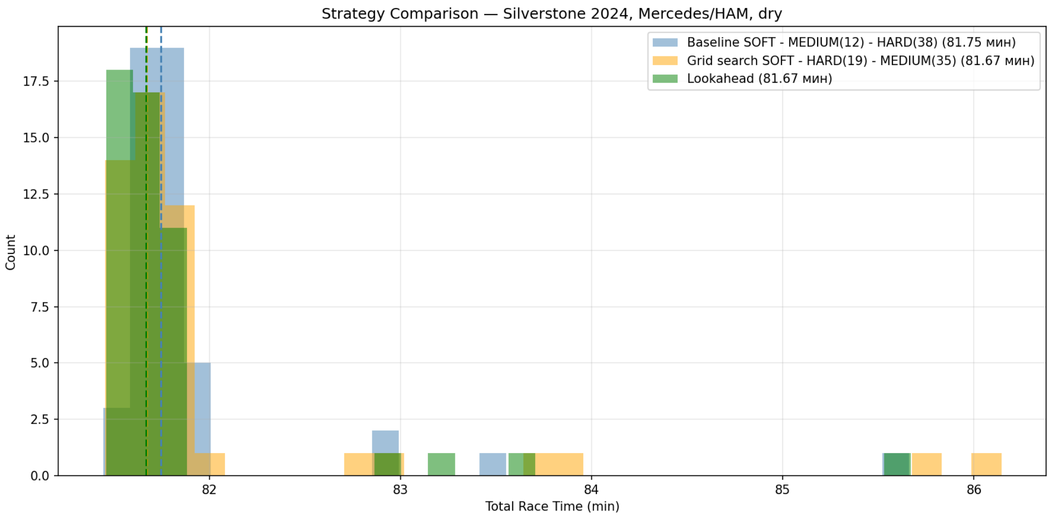

Figure 8.1: Distribution of total race times across 50 Monte Carlo simulations for three pit stop strategies - Silverstone 2024, Mercedes/Hamilton, dry conditions. Dashed lines indicate median race time for each strategy.

8.2 Race with 3 pit-stop

As a second test case, Interlagos was selected - the only circuit in the dataset with a median of 3 pit stops per driver per race. The race consists of 69 laps, requiring a fundamentally different strategic approach compared to Silverstone.

50 Monte Carlo simulations were run for each of the three strategies. The baseline makes 3 pit stops based on historical thresholds - SOFT → MEDIUM(17) → HARD(36) → MEDIUM(61). Grid search found an aggressive strategy with early pit stops - SOFT → MEDIUM(10) → SOFT(18) → HARD(46). Lookahead made decisions independently at each lap.

The results differ fundamentally from Silverstone - the distributions are substantially wider and overlap considerably, with race times spanning 92–104 minutes compared to 81–86 minutes at Silverstone. This reflects the greater role of stochastic events in a long race with three pit stops. Median times: Baseline - 96.36 min, Grid search - 95.93 min, Lookahead - 95.94 min. Grid search and Lookahead improve upon the baseline by approximately 25 seconds - a more substantial gain than at Silverstone, suggesting greater optimisation potential on circuits with a higher number of pit stops.

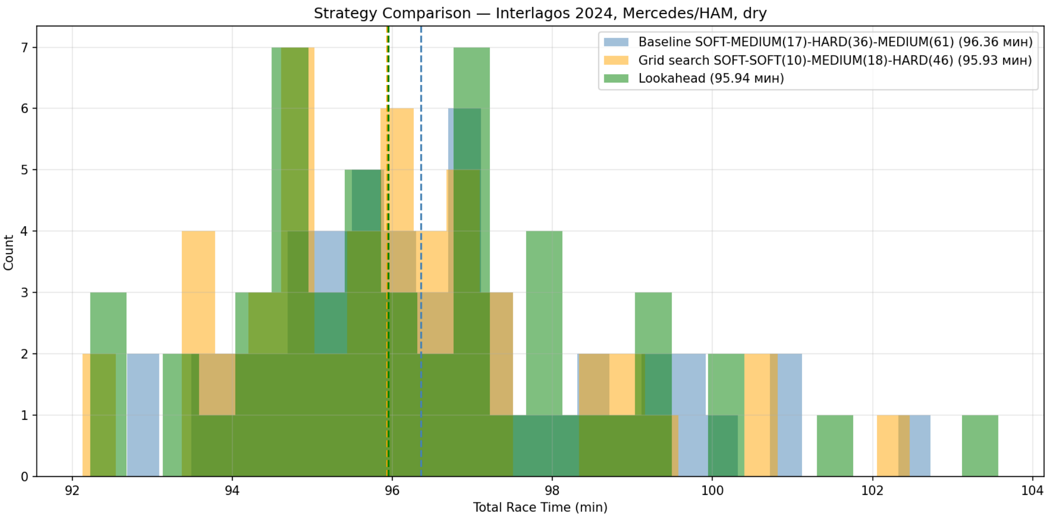

Figure 8.2: Distribution of total race times across 50 Monte Carlo simulations for three pit stop strategies - Interlagos 2024, Mercedes/Hamilton, dry conditions. Dashed lines indicate median race time for each strategy.

8.3 Safety Car Scenario Analysis

To evaluate strategy robustness under stochastic race events, a scenario analysis was conducted at Interlagos (69 laps) with a forced Safety Car deployment at different stages of the race. Early SC - lap 10 (∼14% of race distance), mid SC - lap 30 (∼43%), late SC - lap 50 (∼72%). Each scenario was run 50 times for each of the three algorithms.

Table 8.1: Safety Car scenario analysis - Interlagos 2024, Mercedes/HAM, dry conditions. Median total race time (minutes) across 50 Monte Carlo simulations per scenario.

Scenario	Baseline	Grid Search	Lookahead
No safety-car	96.355	95.919	95.944
Early SC	98.046	97.614	97.895
Mid SC	98.131	98.060	97.976
Late SC	98.248	97.948	98.315

Grid search consistently improves upon the baseline across all scenarios - by 4 to 26 seconds. The smallest advantage under mid-race SC is explained by the fact that SC disrupts the planned stints, forcing either short stints or running beyond the optimal tyre life. This is precisely the scenario where Lookahead outperforms grid search (-9.3 s vs. -4.2 s relative to baseline) - adapting to SC in real time, it makes a more favourable pit decision based on the current tyre state, whereas grid search follows a fixed plan that SC disrupts.

Lookahead shows mixed results in the remaining scenarios. Under early SC (lap 10) the algorithm loses its advantage - the tyres are still too fresh for a pit stop under SC to be beneficial, yet the algorithm pits anyway. Under late SC (lap 50) Lookahead marginally underperforms the baseline - the remaining laps are insufficient to recover the cost of an additional pit stop.

9 Conclusion

This work presents the development and implementation of a simulation environment for evaluating and comparing pit stop strategies in motorsport, based on the analysis of telemetry and historical Formula 1 data spanning the 2011–2024 seasons.

The study included a comprehensive data analysis that confirmed several key patterns: the dependence of pit stop frequency and structure on circuit characteristics, the nonlinear nature of tyre degradation with diminishing marginal lap time loss, a substantial reduction in positional losses when pitting under Safety Car conditions, and the relationship between pit stop timing and the resulting position change. Of the five hypotheses formulated, four were confirmed statistically (H1, H2, H3, H5), while hypothesis H4 concerning the influence of team operational efficiency did not reach statistical significance.

The simulation environment follows a modular architecture and comprises five components: a base pace model (XGBoost, MAE = 5.3 s), a track evolution model (logarithmic function, R^2 = 0.984), per-compound tyre degradation models (MAE 0.5 s/lap), a probabilistic Safety Car model, and a pit stop duration model (MAE = 1.7 s).

Three strategies were implemented and evaluated within the environment: a rule-based baseline, a grid search with Monte Carlo evaluation, and an adaptive Lookahead policy. Testing on two circuits showed that both grid search and the Lookahead policy consistently improve upon the baseline by 4–26 seconds, with the advantage more pronounced on circuits with three pit stops. Safety Car scenario analysis demonstrated that the Lookahead policy outperforms grid search under a mid-race Safety Car by adapting in real time, but underperforms under early or late Safety Car deployments due to the limitations of its planning horizon.

Overall, the proposed approach confirms the applicability of optimisation and simulation methods for supporting strategic decision-making in motorsport.

10 Directions for Future Research

- **Extending the simulation environment.** The current environment models a single participant without explicit interaction with competitors. Incorporating multi-agent dynamics - simulating the actions of multiple drivers simultaneously while accounting for traffic and undercuts - will bring the simulation closer to real racing conditions and open up the possibility of applying game-theoretic methods discussed in the literature review.

- **Improving the tyre degradation model.** The current models are trained on aggregated historical data and do not account for individual driving style or the thermal regimes of a specific car. Integrating high-frequency telemetry data and Bayesian state-update methods during the race will enable more accurate real-time estimates of remaining tyre life.

- **Reinforcement learning.** The Lookahead policy is constrained by a fixed 20-lap horizon and an analytical cost function. A promising direction is to replace or supplement this policy with a reinforcement learning agent trained directly within the simulation environment - analogous to the approach described in Thomas et al. (2025). This would yield strategies that adapt to arbitrary race scenarios without manually specifying a cost function.

- **Modelling weather transitions.** In the current implementation, weather conditions are fixed at race start. Modelling dynamic weather changes during the race - most importantly the dry-to-wet transition - and the associated selection of intermediate and wet tyres would substantially increase the realism of strategic scenarios.

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Appendix: AI Prompts and Responses

Prompt 1:

"I have 12 papers on pit stop strategy optimization in Formula 1, attaching the list with references. For each paper, describe the methods and algorithms used by the authors in their work."

Answer:

1 **Boettinger & Klotz (2023)** — Physics-based race simulation, discrete-event modeling of race processes, reinforcement learning methods for strategy evaluation and optimization.

2 **Todd et al. (2025)** — Time series forecasting for tire energy dynamics estimation, explainable machine learning methods with feature attribution mechanisms.

3 **Aguad & Thraves (2024)** — Dynamic programming for sequential decision-making, zero-sum feedback Stackelberg game formulation, backward induction for equilibrium computation.

4 **van Kampen et al. (2024)** — Model Predictive Control (MPC), dynamic model of an electric race car with energy and battery constraints, constrained optimal control problem formulation.

5 **Cappello & Hoegh (2025)** — Bayesian state-space model for tire degradation, latent state inference from lap time data, skewed-t observation model for asymmetric error distribution.

6 **Carrasco Heine & Thraves (2022)** — Discrete-time dynamic programming, backward induction, cost function formulation based on lap time and pit stop time loss.

7 **Thomas et al. (2025)** — Reinforcement learning with recurrent neural network for policy approximation, explainable ML methods for agent decision interpretation.

8 **Fieni et al. (2025) — Towards Learning-Based** — Multi-level optimization, combination of machine learning with optimization models, modular strategic layer architecture.

9 **Fieni et al. (2025) — Game Theory** — Multi-agent dynamic game formulation, physics-based models of vehicle dynamics and tire degradation, equilibrium concepts for stable strategy analysis.

10 **Sasikumar et al. (2025)** — Deep learning for pit stop timing classification and prediction, supervised learning on historical Formula 1 race data.

11 **Heilmeier et al. (2020)** — Artificial neural network as a strategy function approximator, supervised learning on race simulator data, real-time inference architecture via single forward pass.

12 **Bekker & Lotz (2009)** — Discrete-event simulation, probabilistic models of lap times and event durations, strategy comparison through repeated simulation runs.

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Prompt 2

"I have Formula 1 race data. For each race, I want to retrieve weather conditions (temperature, humidity, wind speed, precipitation) at the moment of race start. I have data on the date and start time of each race, the city and country name, and the GPS coordinates of the starting location. Please provide a step-by-step algorithm for how I can obtain the required data."

Answer:

Step 1: Choose a weather data source. The best free option for historical weather data with hourly resolution is Open-Meteo (open-meteo.com) — it supports historical data, requires no API key, and accepts coordinates directly.

Step 2: Prepare your data. Make sure your dataset has the following columns:

- `latitude`, `longitude` — GPS coordinates,
- `date` — race date in `YYYY-MM-DD` format,
- `start_time` — race start time (ideally in UTC).

Step 3: Build the API request. Open-Meteo historical endpoint:

```
https://archive-api.open-meteo.com/v1/archive?
  latitude={lat}
  &longitude={lon}
  &start_date={date}
  &end_date={date}
  &hourly=temperature_2m,relativehumidity_2m,
          precipitation,windspeed_10m
  &timezone=UTC
```

Step 4: Extract the right hour. The API returns hourly data for the whole day. You need to match the race start hour to the correct index in the response array.

```
import requests

def get_weather(lat, lon, date, hour):
    url = "https://archive-api.open-meteo.com/v1/archive"
    params = {
        "latitude": lat,
        "longitude": lon,
        "start_date": date,
        "end_date": date,
        "hourly": "temperature_2m,relativehumidity_2m,"
                  "precipitation,windspeed_10m",
        "timezone": "UTC"
    }
    response = requests.get(url, params=params).json()
    hourly = response["hourly"]
    return {
        "temperature": hourly["temperature_2m"][hour],
        "humidity":    hourly["relativehumidity_2m"][hour],
        "precipitation": hourly["precipitation"][hour],
        "wind_speed":  hourly["windspeed_10m"][hour]
    }
```

Step 5: Loop over all races.

```
import pandas as pd

df = pd.read_csv("races.csv")
df["start_hour"] = pd.to_datetime(df["start_time"]).dt.hour

results = []
for _, row in df.iterrows():
    weather = get_weather(
        lat=row["latitude"],
        lon=row["longitude"],
        date=row["date"],
```

```
        hour=row["start_hour"]
    )
    results.append(weather)

weather_df = pd.DataFrame(results)
final_df = pd.concat([df, weather_df], axis=1)
final_df.to_csv("races_with_weather.csv", index=False)
```

Prompt 3

"I am developing a simulation environment for F1. My environment consists of several models, including a tyre degradation model and a track evolution model. Help me separate degradation from track evolution - write code to determine the rubbering function."

Answer:

```
from scipy.optimize import curve_fit
from sklearn.metrics import r2_score
import numpy as np
import matplotlib.pyplot as plt

# Empirical points: average pace improvement by lap_frac
# (obtained as the median of early stint laps across all races)
x = np.array([0.0, 0.1, 0.2, 0.3, 0.4, 0.5,
              0.6, 0.7, 0.8, 0.9, 1.0])
y = np.array([3.355, 1.011, 0.750, 0.139, -0.183,
             -0.473, -0.727, -1.155, -1.518, -1.720, -1.640])

# Three candidates
def log_func(x, a, b, c):
    return a * np.log(x + b) + c

coeffs_lin  = np.polyfit(x, y, 1)
coeffs_poly = np.polyfit(x, y, 2)
popt_log, _ = curve_fit(log_func, x, y,
                        p0=[-2, 0.1, 2], maxfev=5000)
```

```
y_lin  = np.polyval(coeffs_lin, x)
y_poly = np.polyval(coeffs_poly, x)
y_log  = log_func(x, *popt_log)

print(f"Linear R2: {r2_score(y, y_lin):.3f}")
print(f"Log    R2: {r2_score(y, y_log):.3f}")
print(f"Poly2  R2: {r2_score(y, y_poly):.3f}")
print(f"Log params: a={popt_log[0]:.3f}, "
      f"b={popt_log[1]:.3f}, c={popt_log[2]:.3f}")

# Visualization
fig, ax = plt.subplots(figsize=(10, 5))
ax.scatter(x, y, color='black', zorder=5,
           s=50, label='Empirical points')
ax.plot(x, y_lin,
        label=f"Linear R2={r2_score(y, y_lin):.3f}",
        linewidth=2)
ax.plot(x, y_log,
        label=f"Log    R2={r2_score(y, y_log):.3f}",
        linewidth=2)
ax.plot(x, y_poly,
        label=f"Poly2  R2={r2_score(y, y_poly):.3f}",
        linewidth=2)
ax.axhline(0, color='gray', linestyle='--', alpha=0.5)
ax.set_xlabel('Lap fraction')
ax.set_ylabel('Rubbering effect (s)')
ax.set_title('Shape of the Track Evolution Curve')
ax.legend()
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```